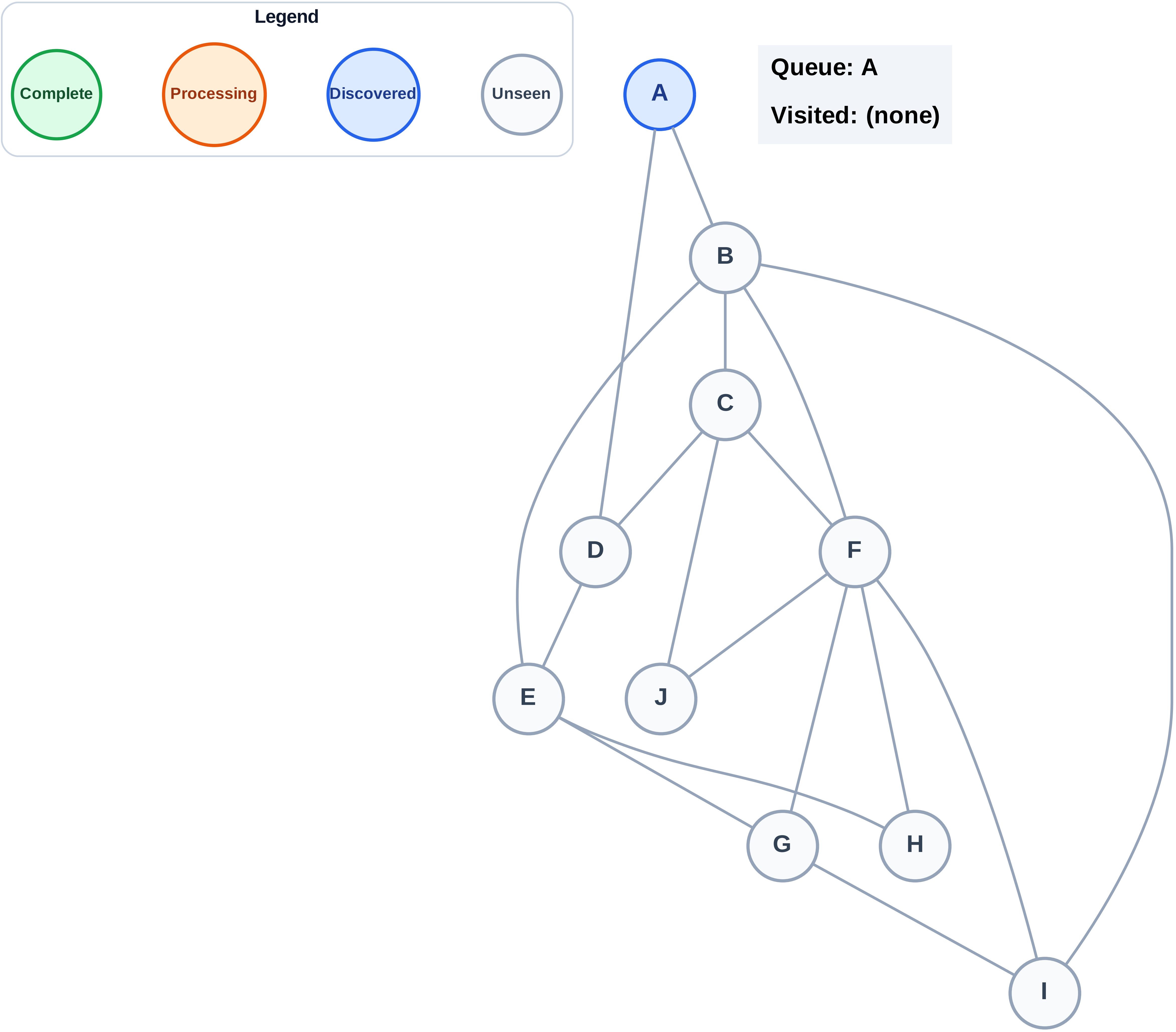
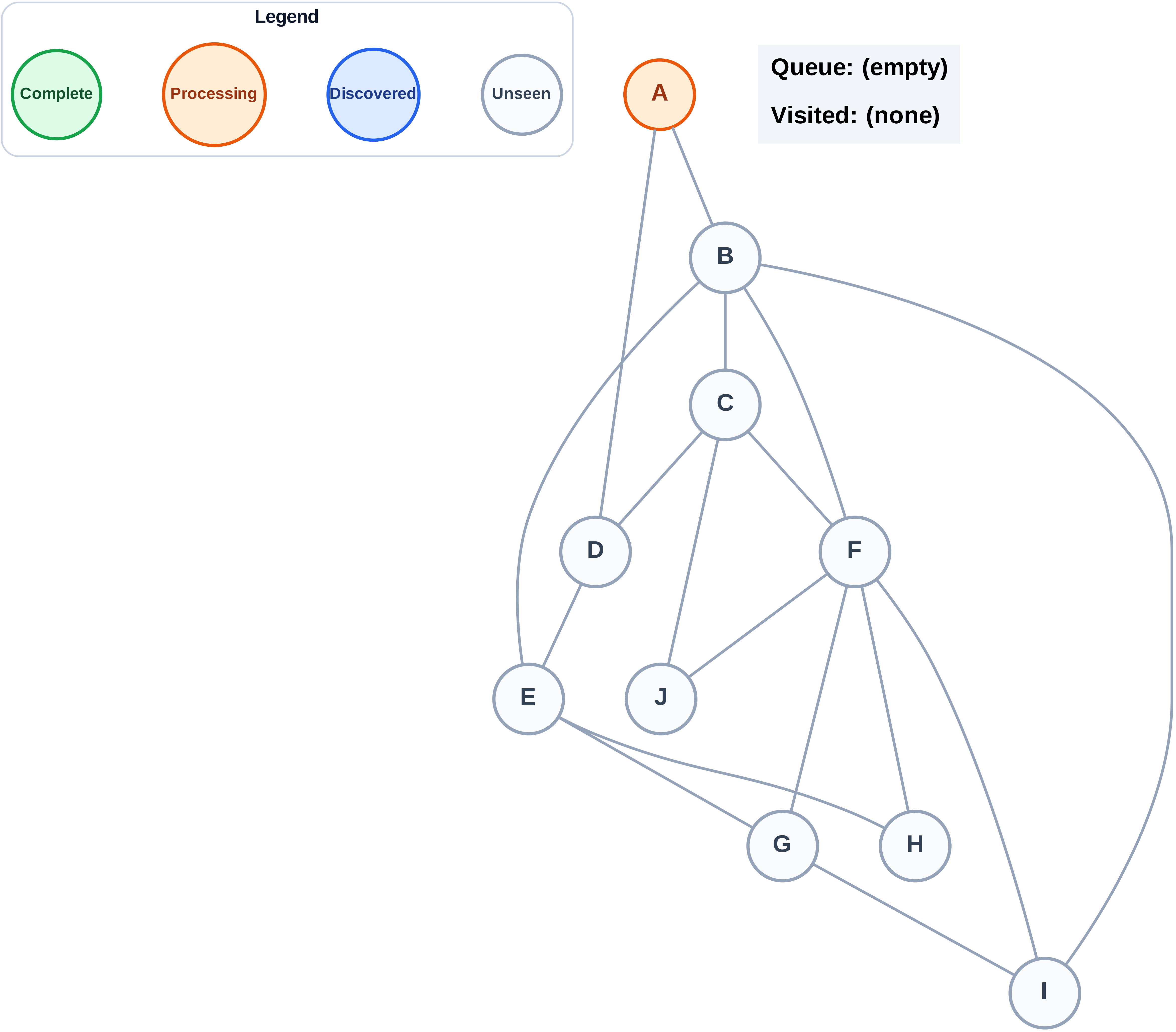
BFS Traversal

Step 1: Start at A



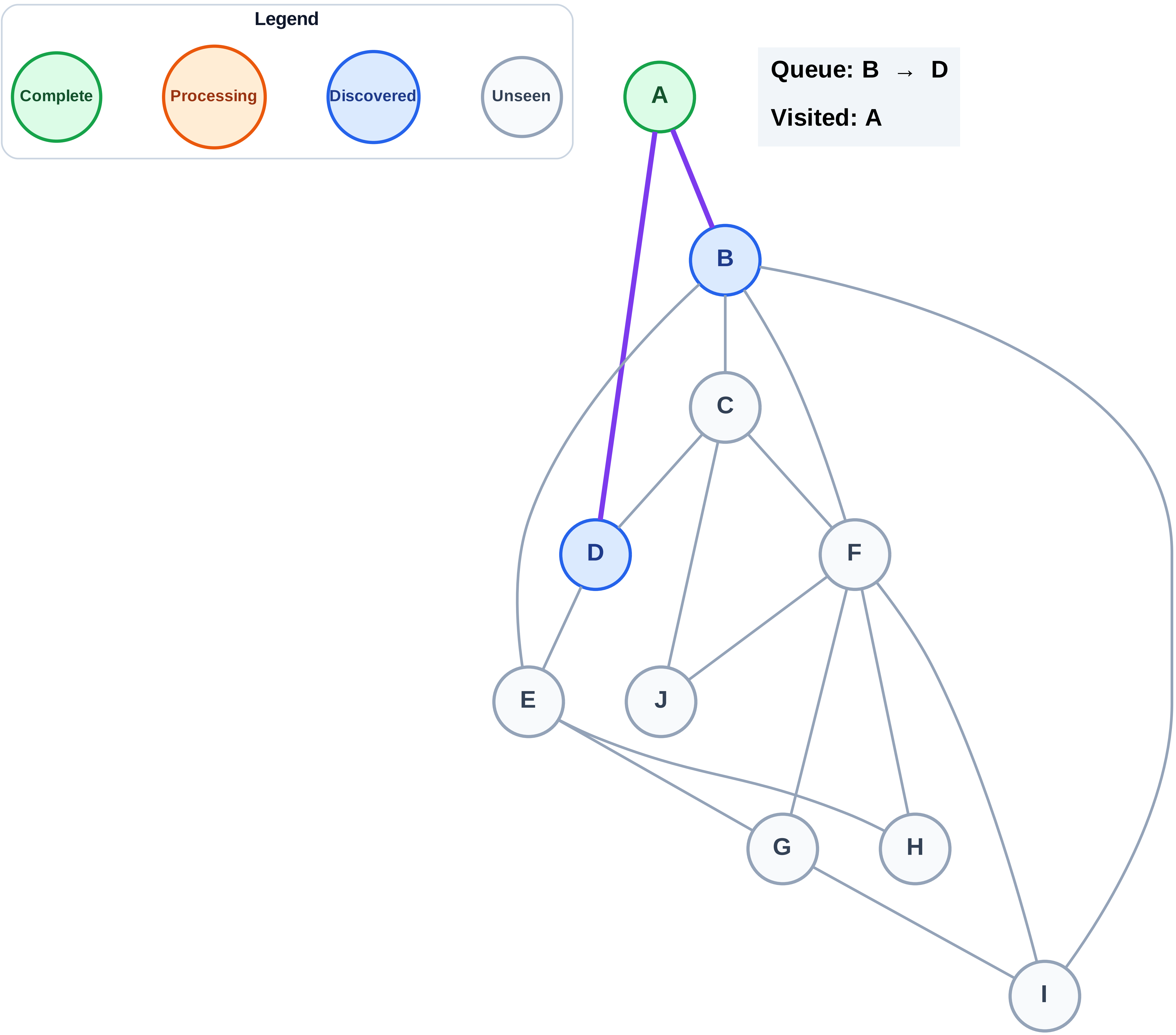
BFS Traversal

Step 2: Process node A



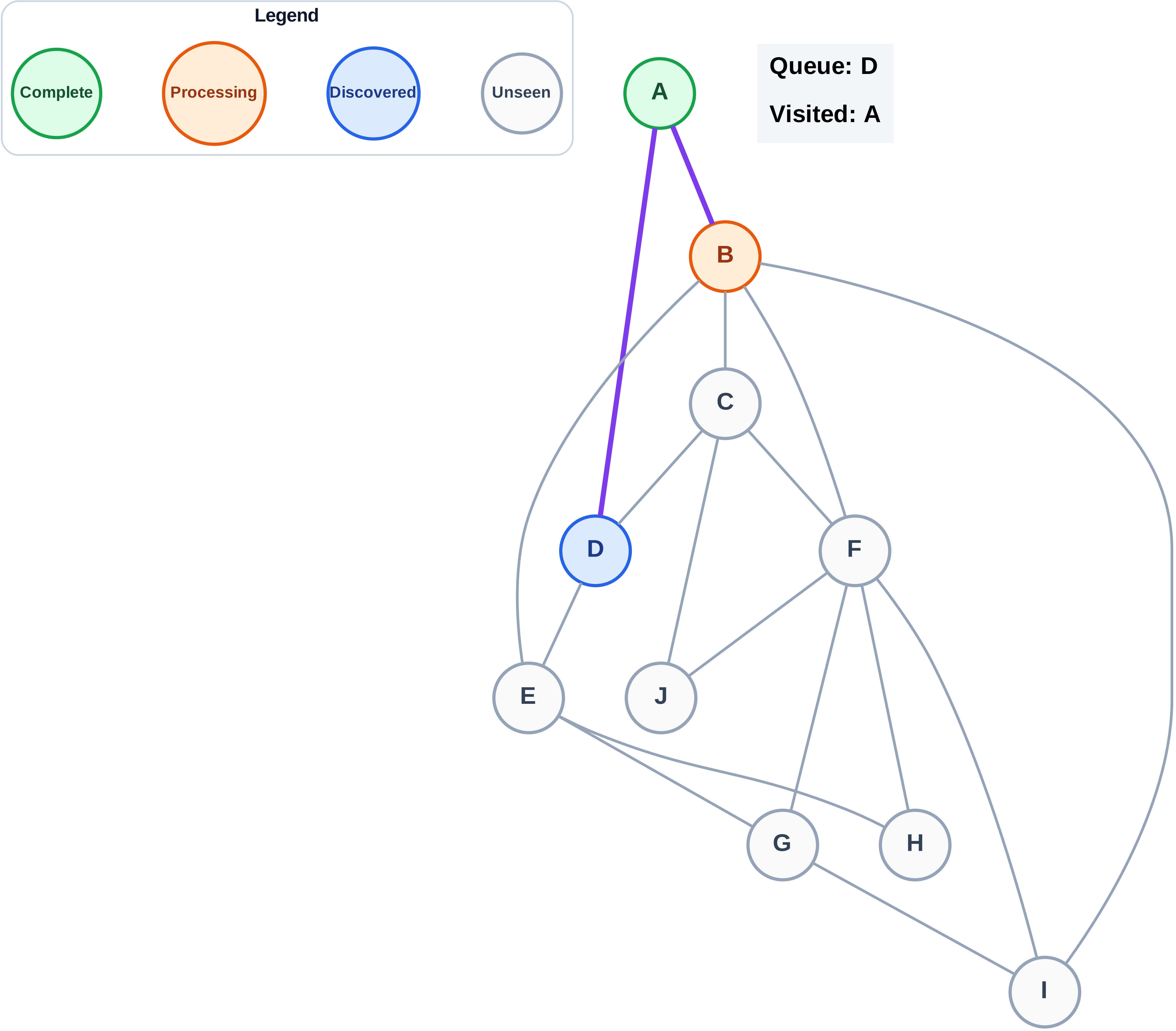
BFS Traversal

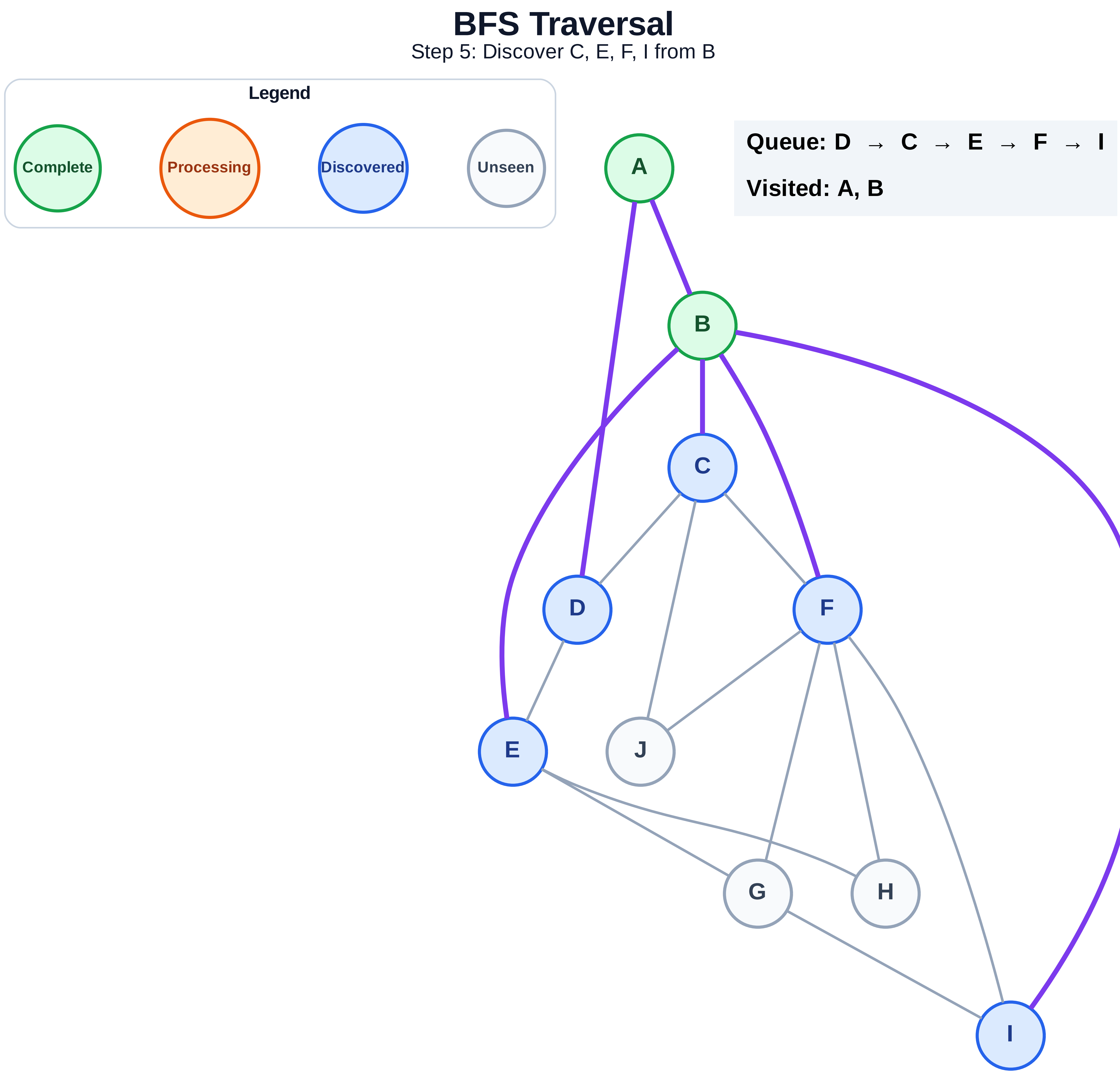
Step 3: Discover B, D from A



BFS Traversal

Step 4: Process node B

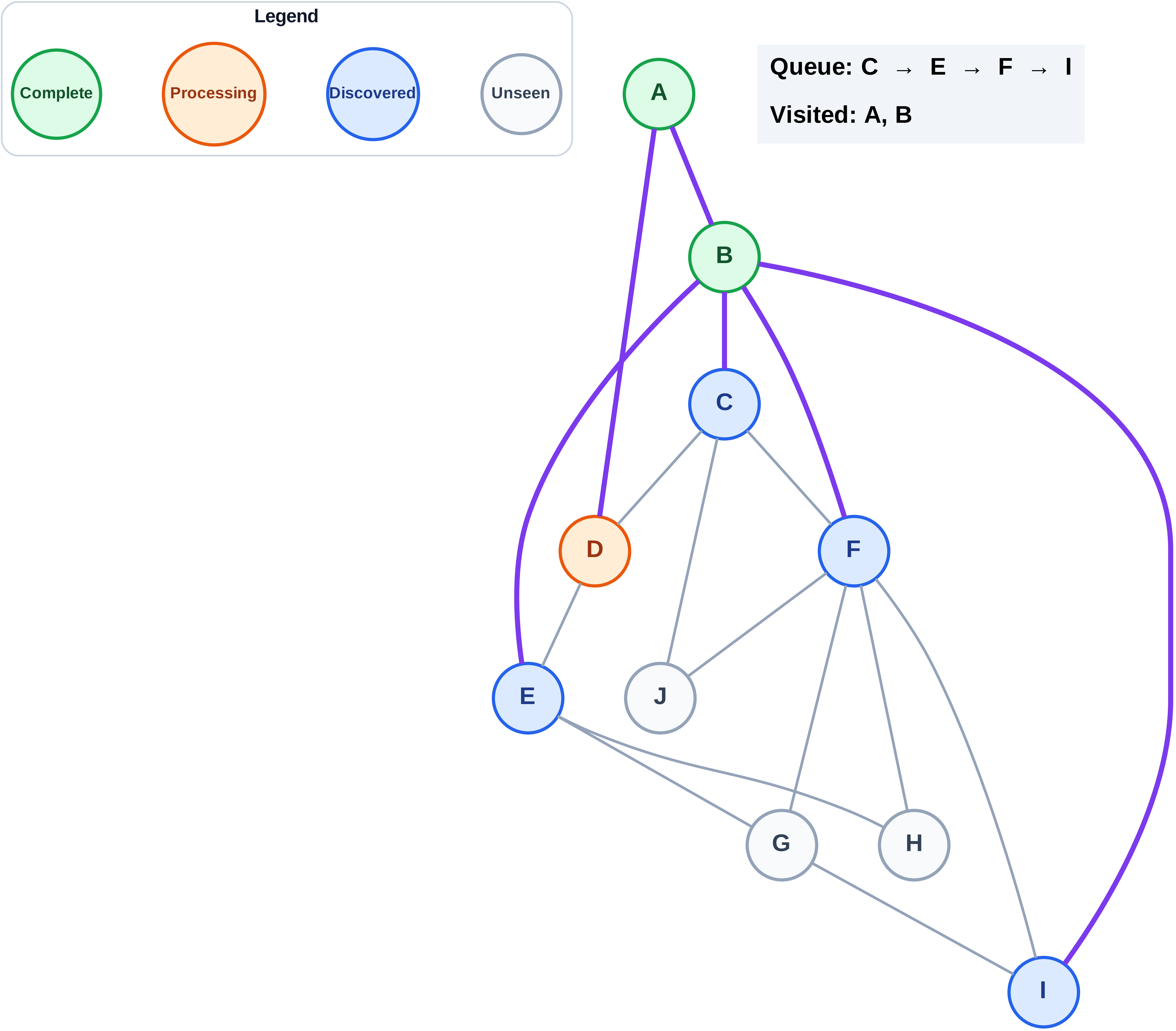




BFS Traversal
Step 5: Discover C, E, F, I from B
Legend
Complete
Processing
Discovered
Unseen
Queue: D → C → E → F → I
Visited: A, B
A
B
C
D
F
E
J
G
H
I

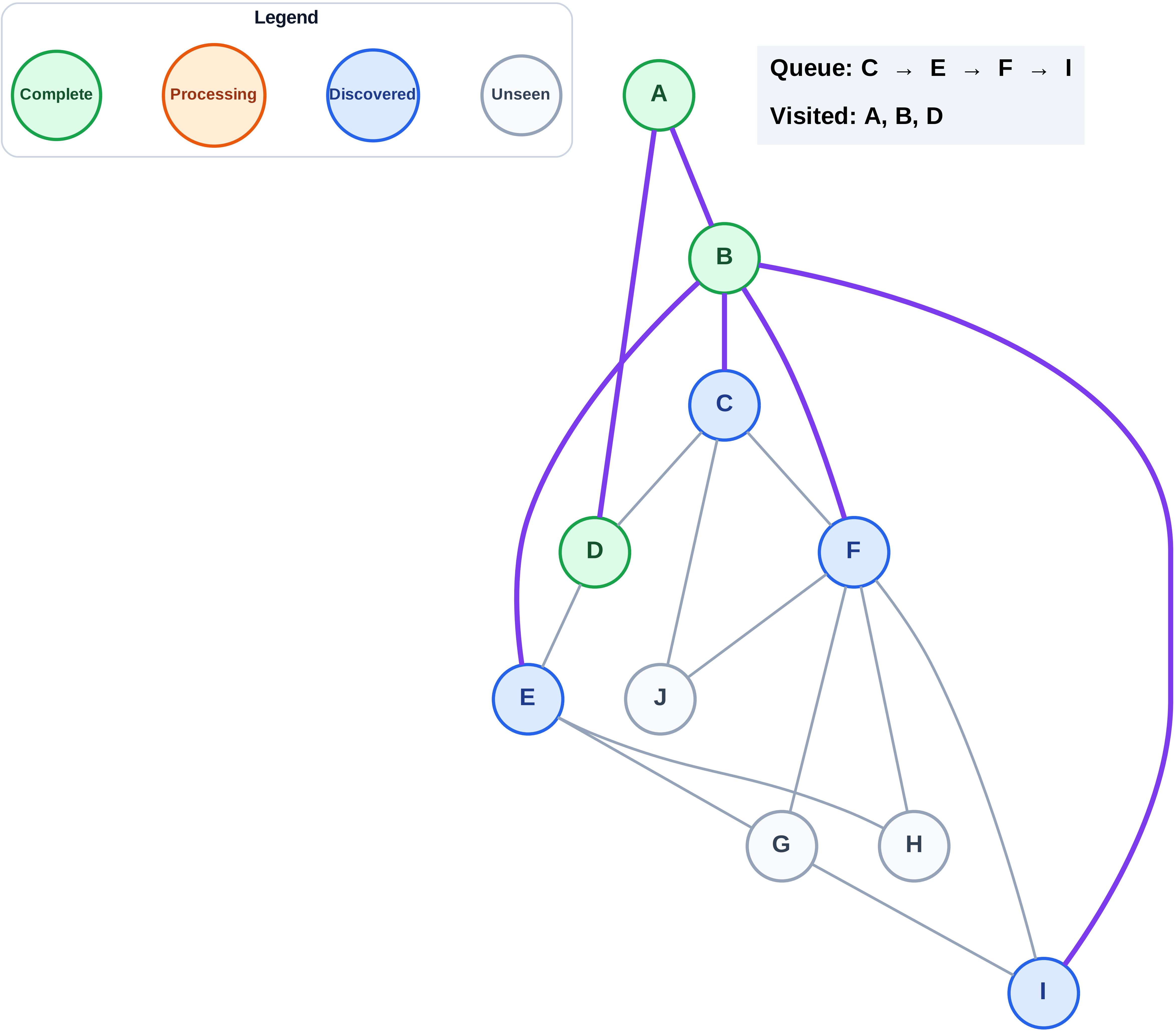
BFS Traversal

Step 6: Process node D



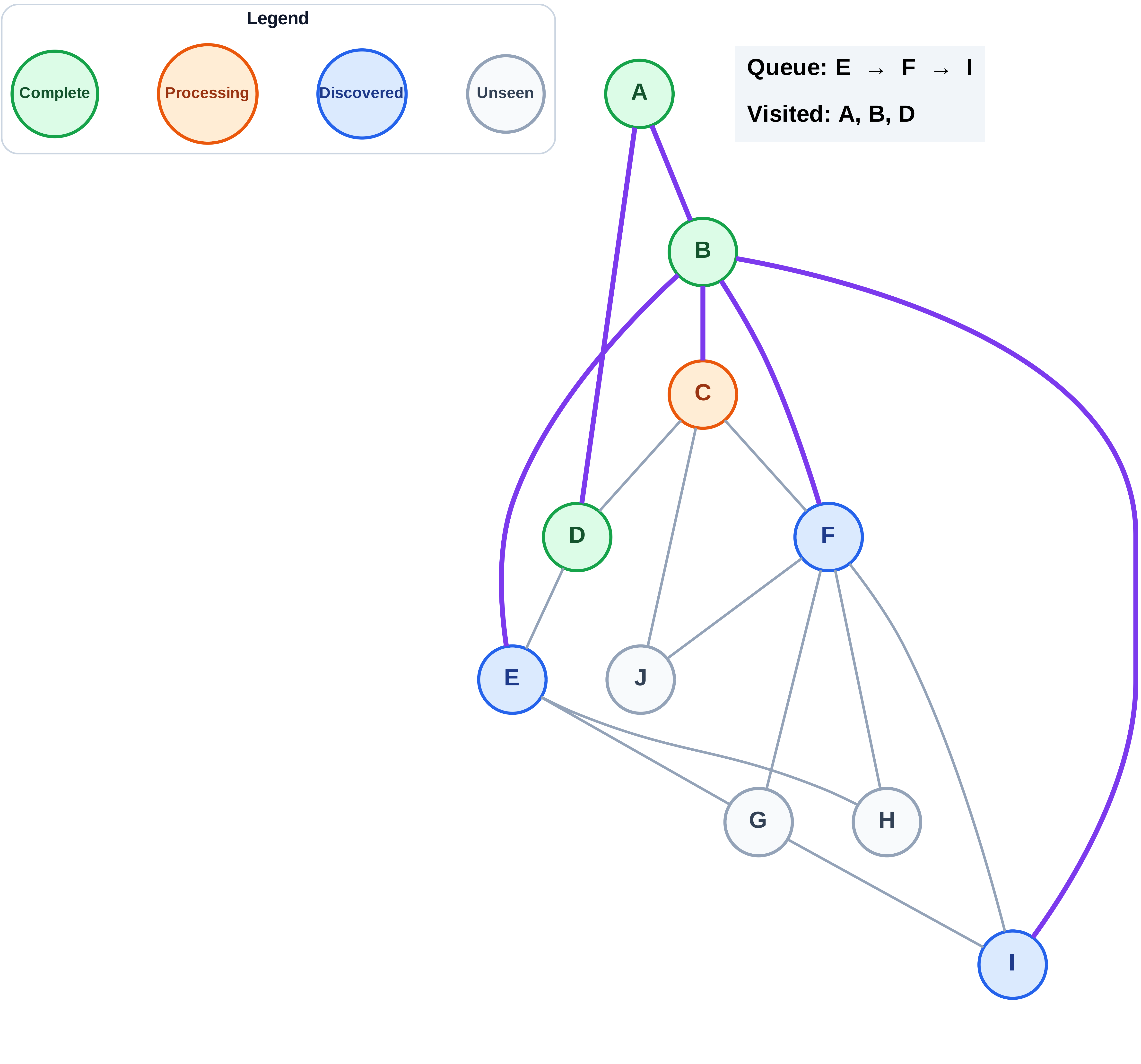
BFS Traversal

Step 7: No new neighbors from D



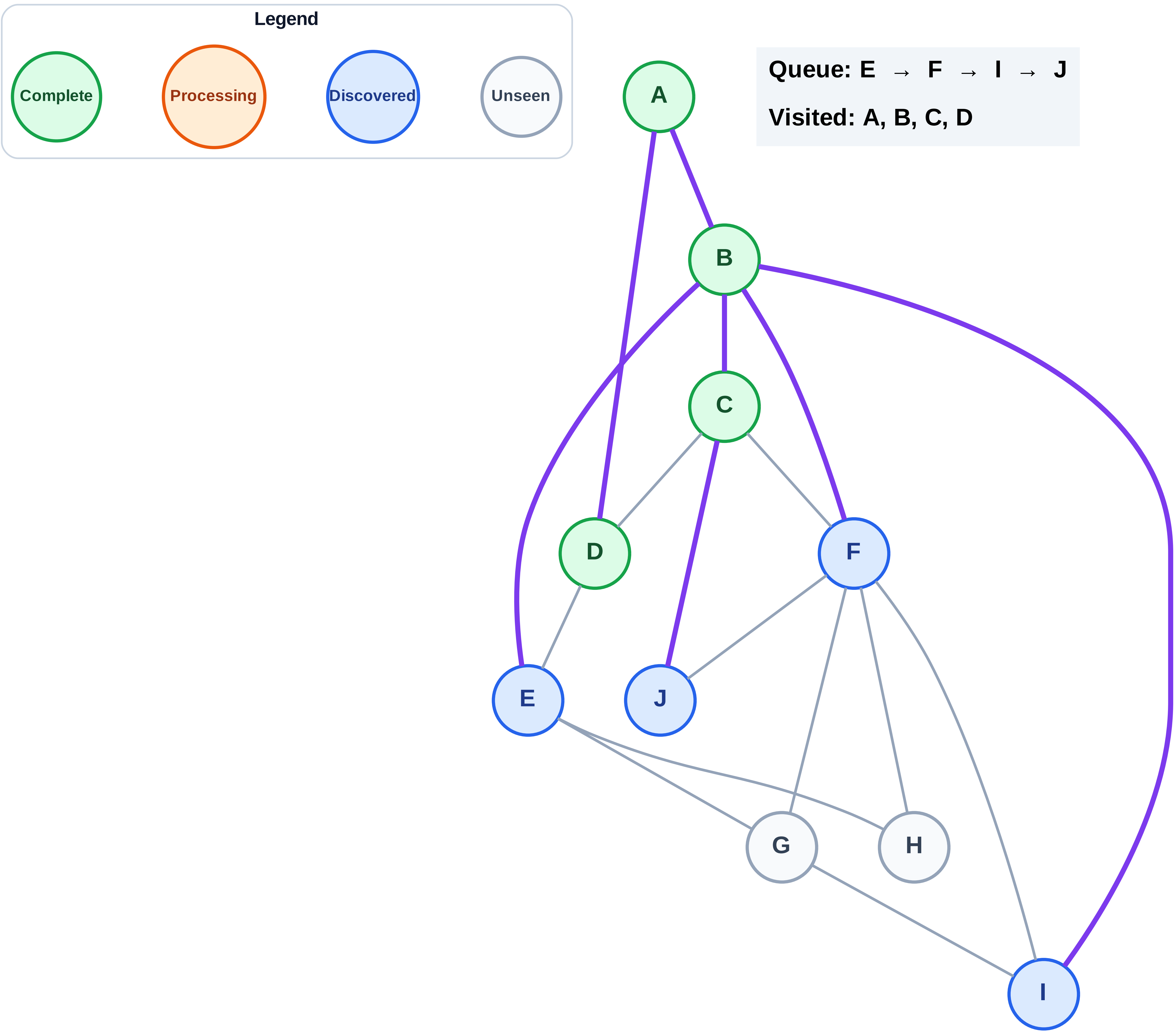
BFS Traversal

Step 8: Process node C



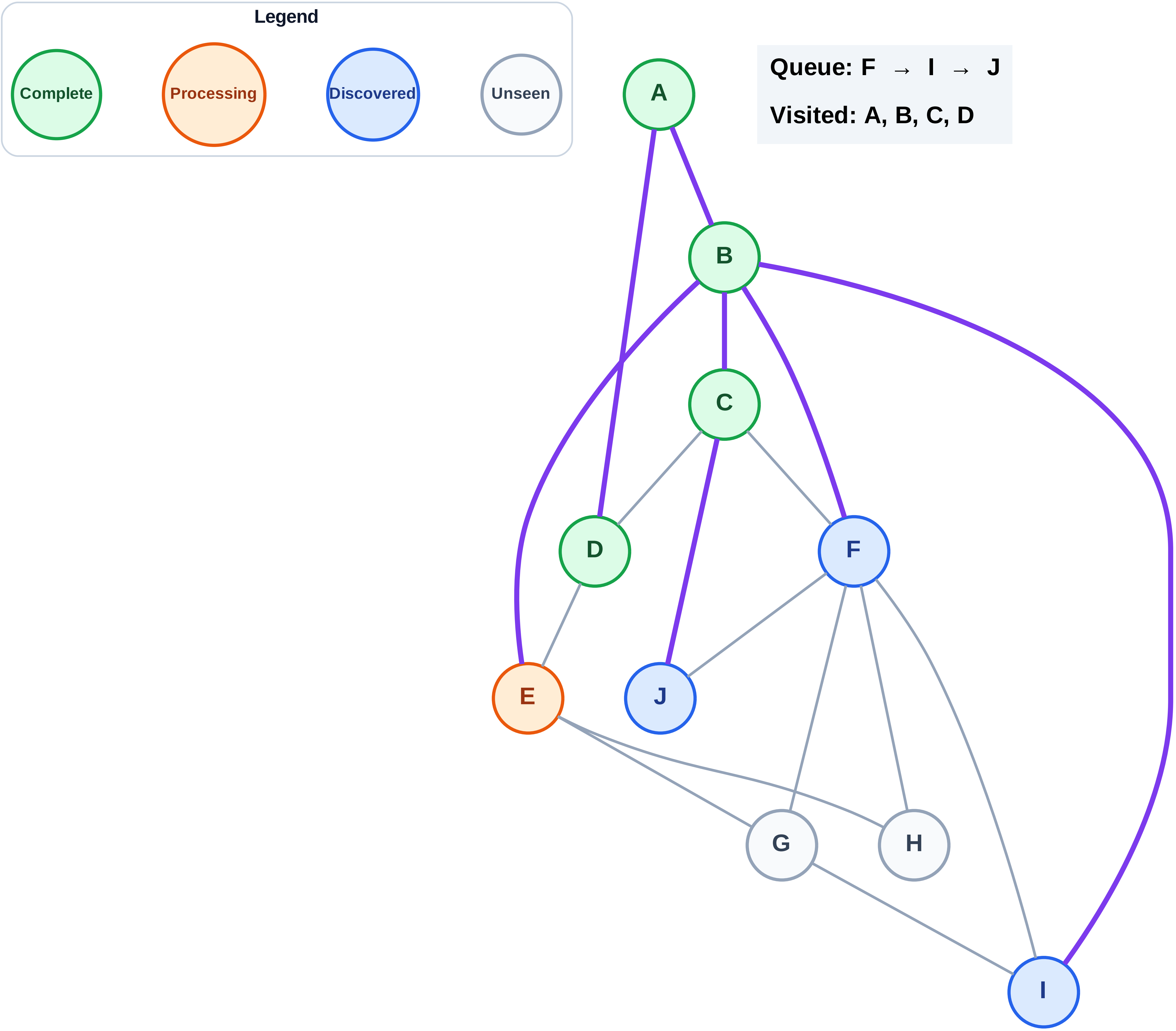
BFS Traversal

Step 9: Discover J from C



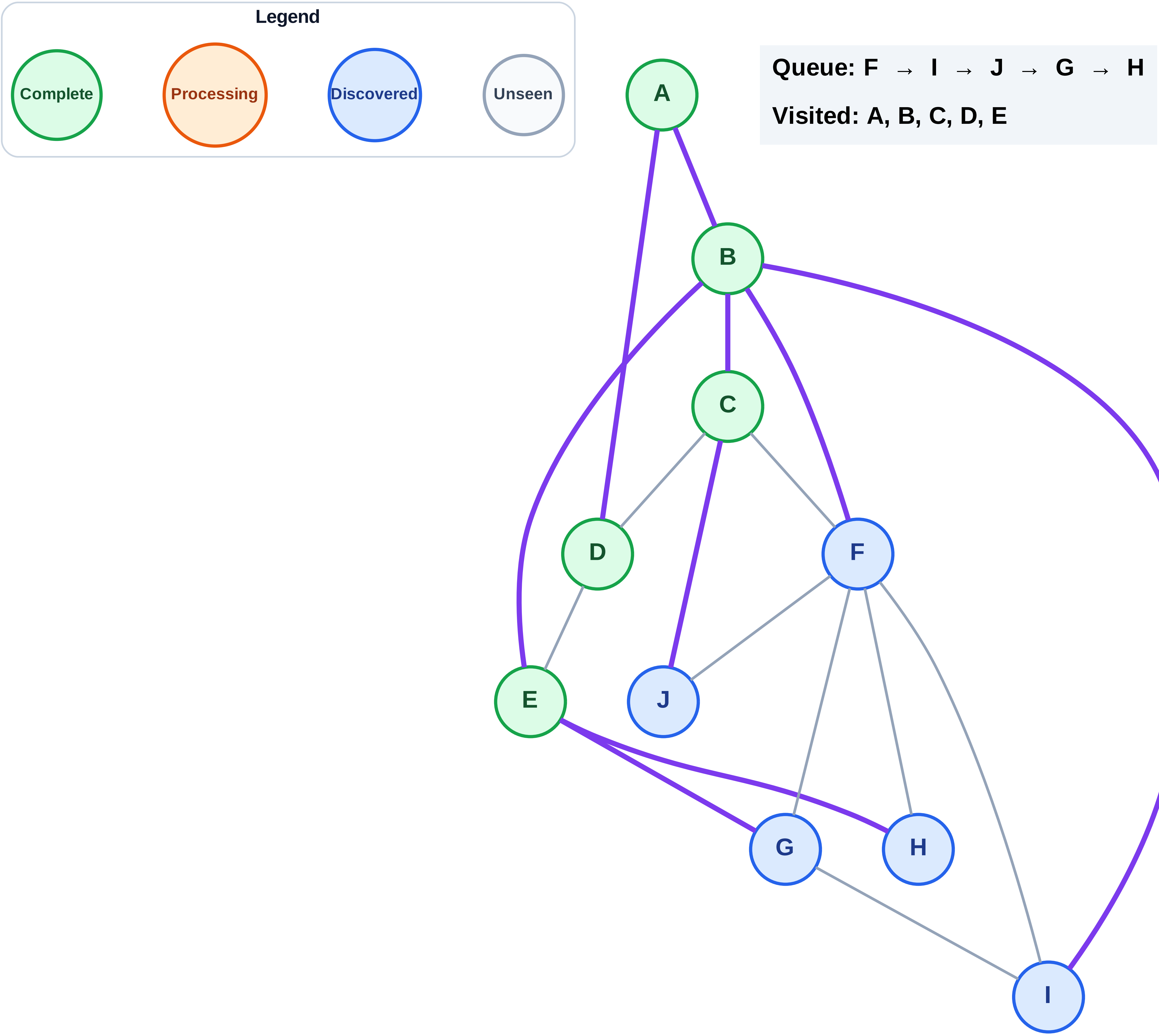
BFS Traversal

Step 10: Process node E



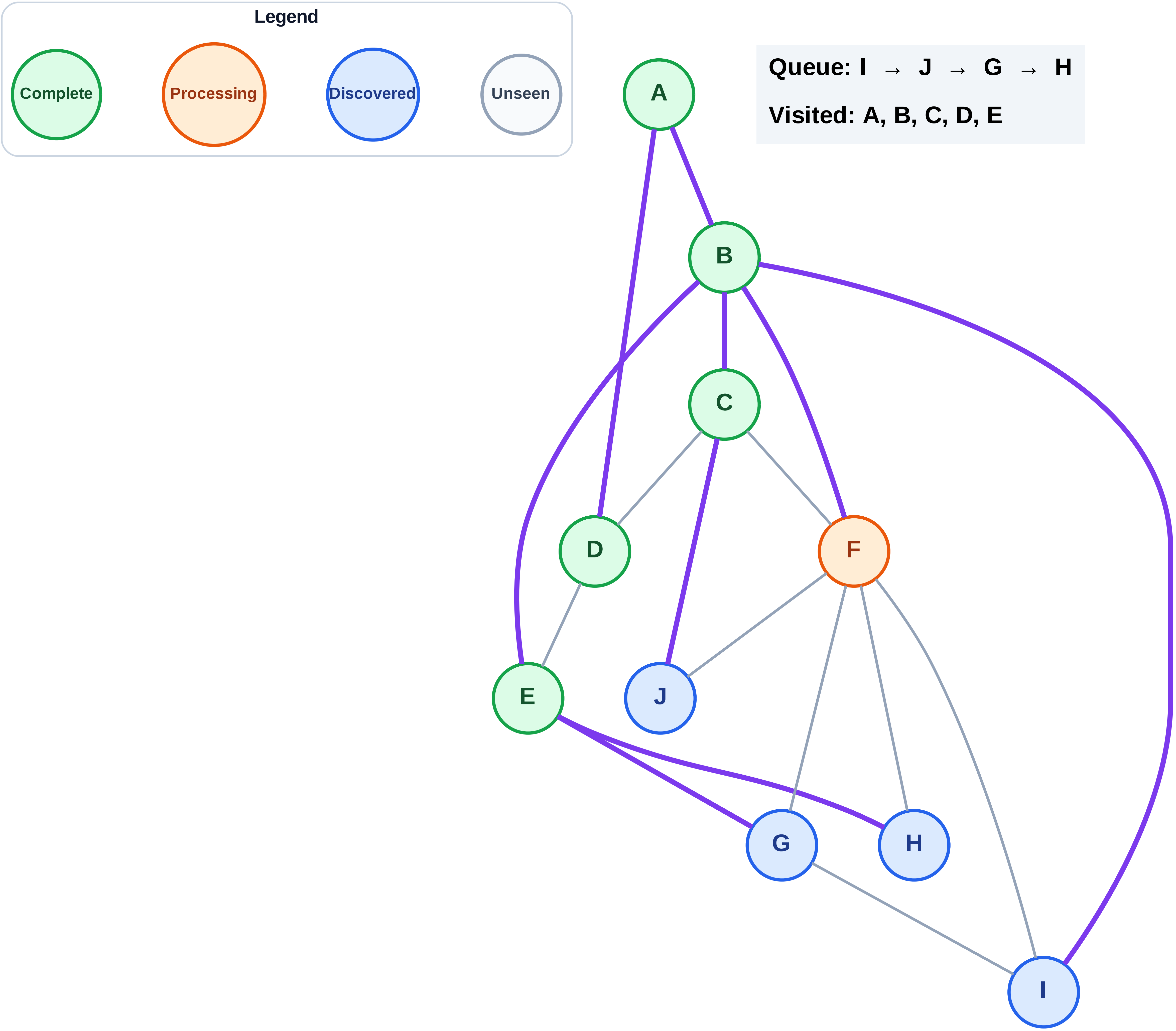
BFS Traversal

Step 11: Discover G, H from E



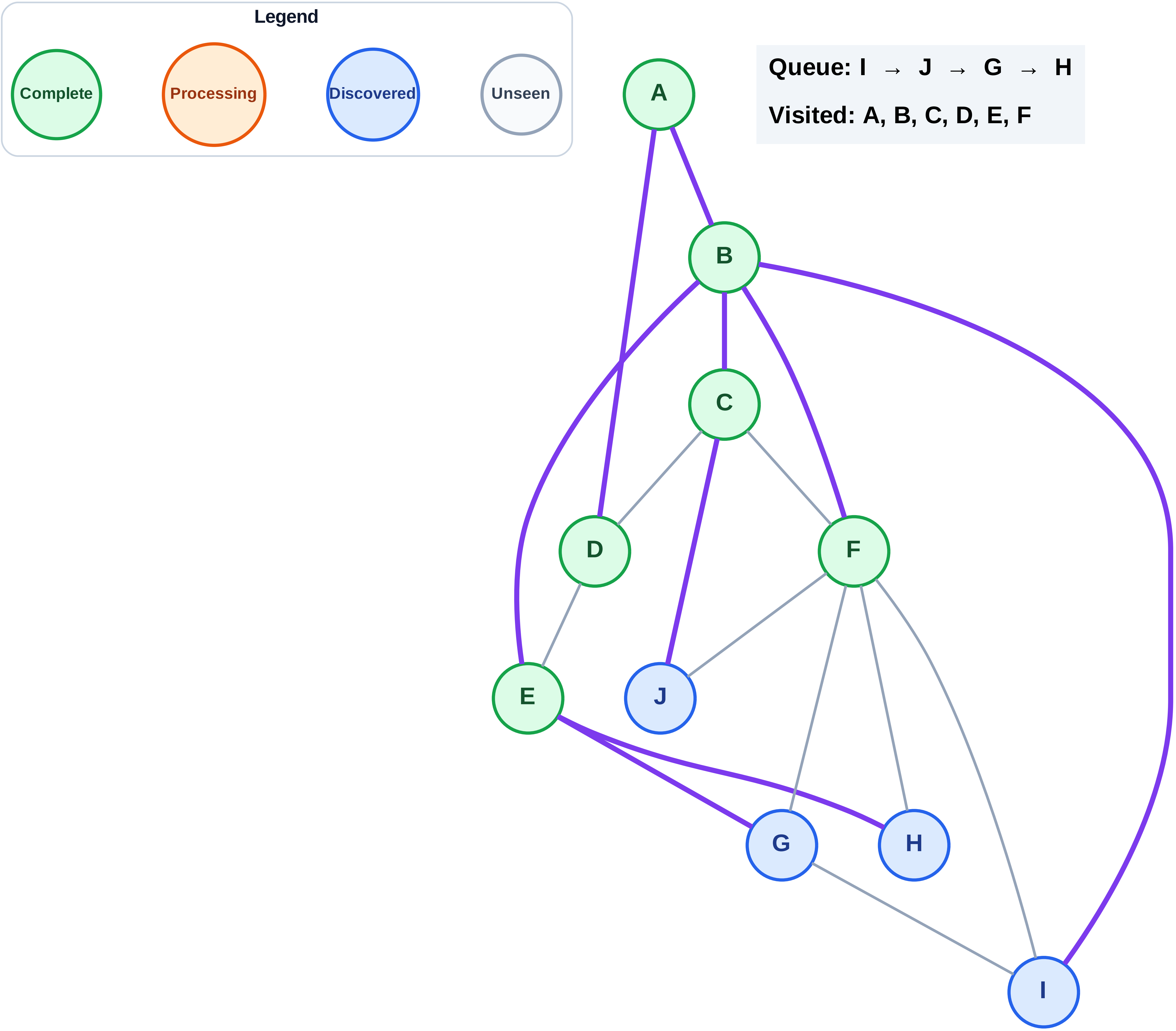
BFS Traversal

Step 12: Process node F



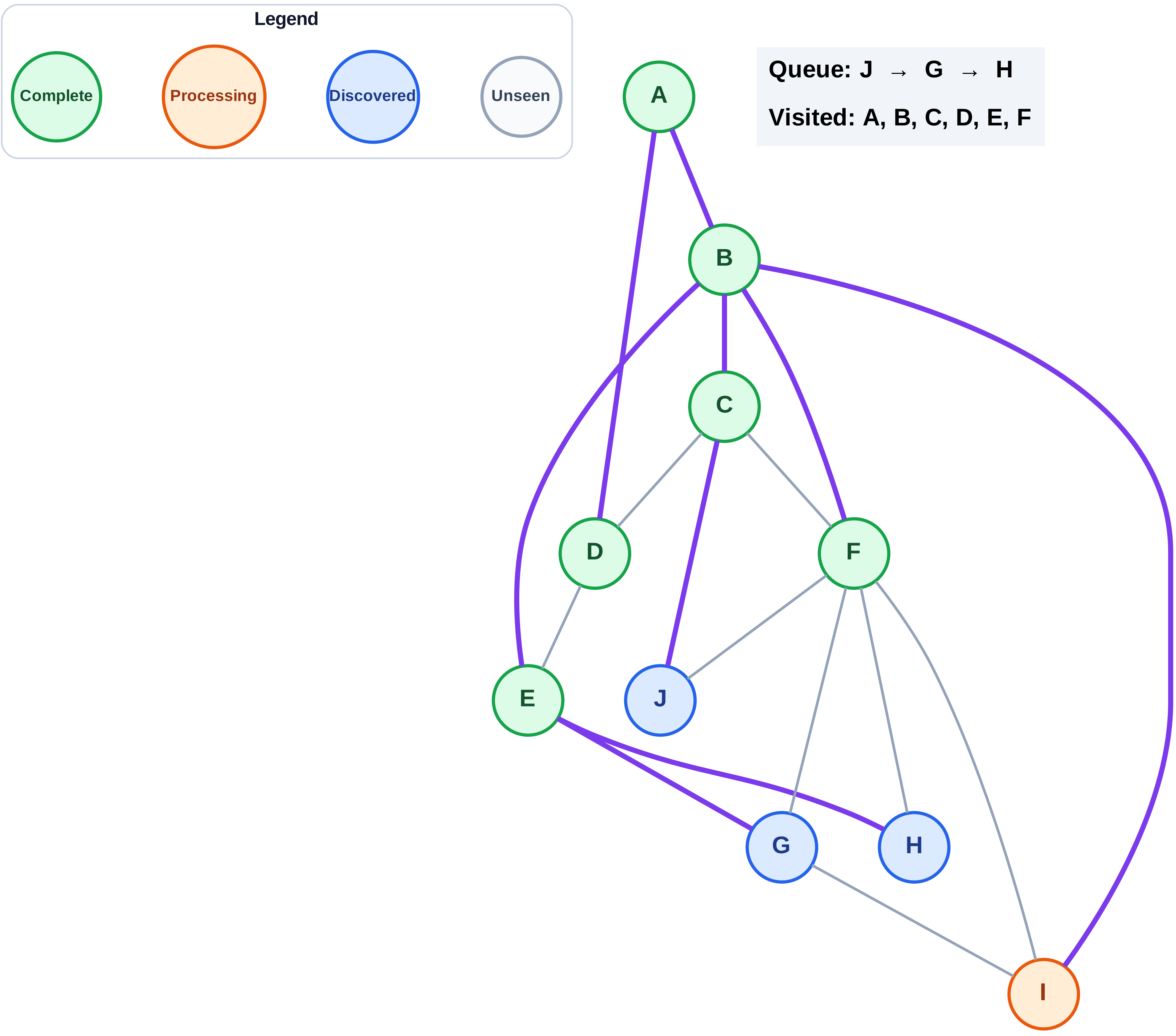
BFS Traversal

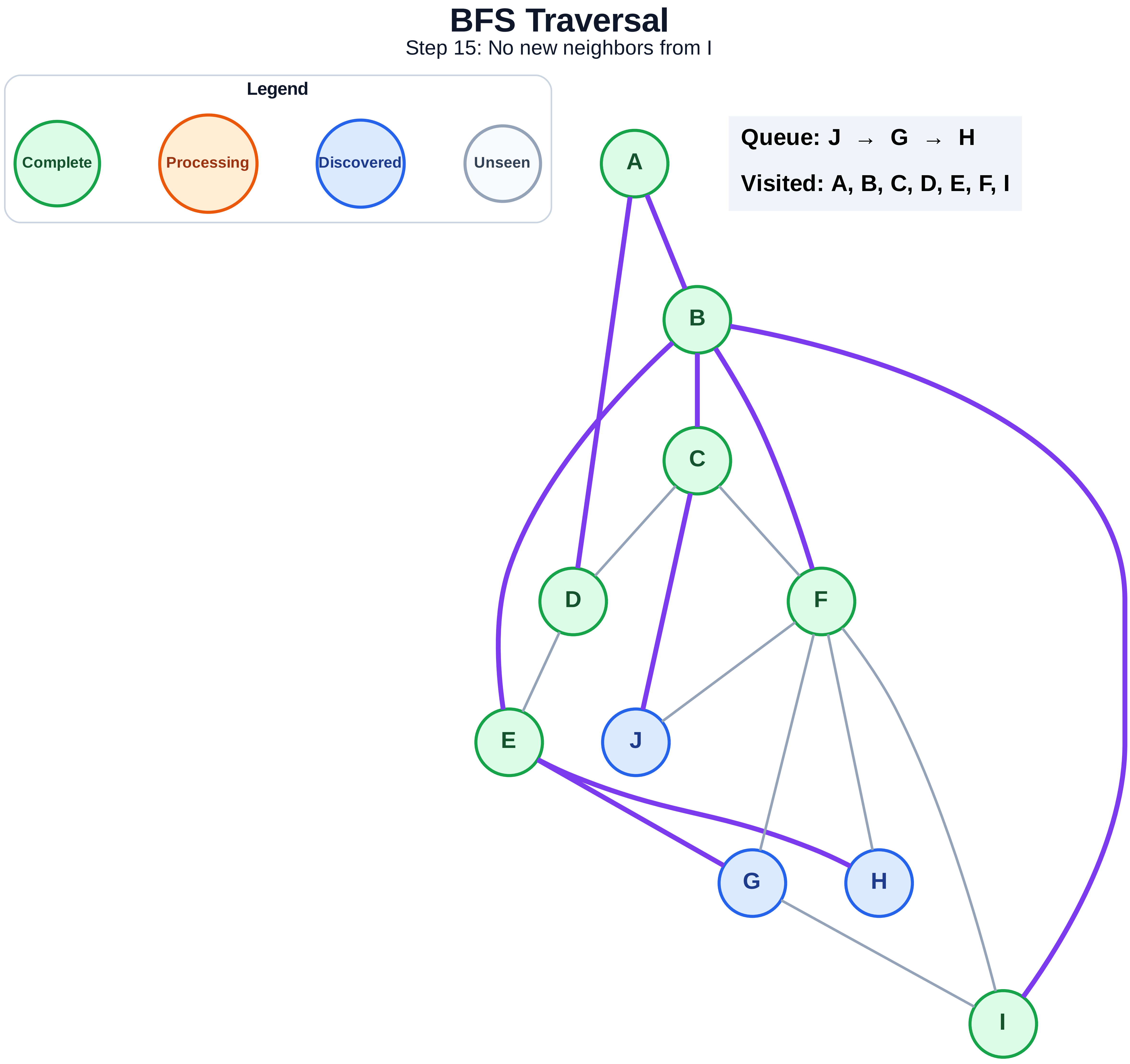
Step 13: No new neighbors from F



BFS Traversal

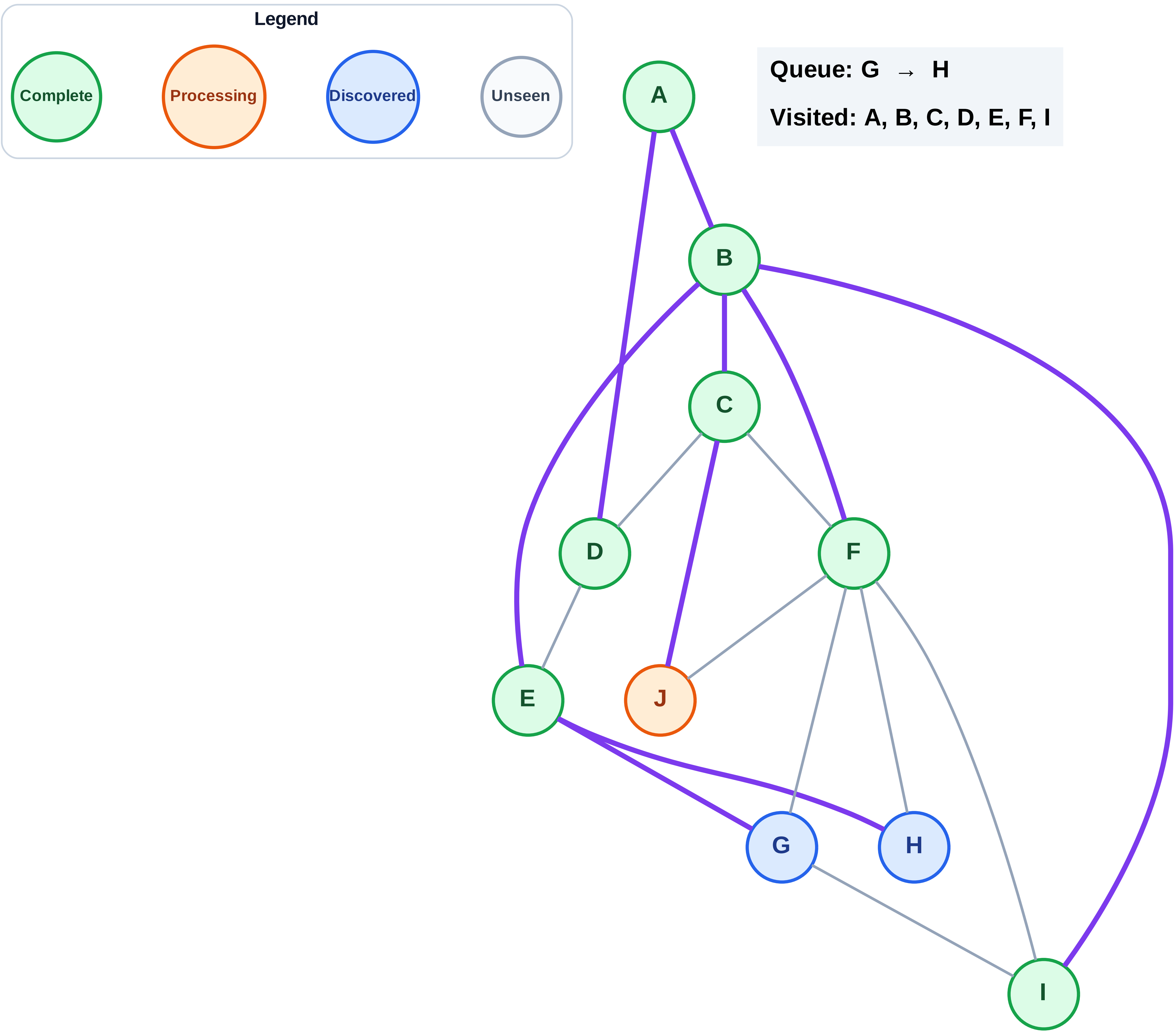
Step 14: Process node I




BFS Traversal
Step 15: No new neighbors from I
Legend
Complete
Processing
Discovered
Unseen
Queue: J → G → H
Visited: A, B, C, D, E, F, I
A
B
C
D
F
E
J
G
H
I

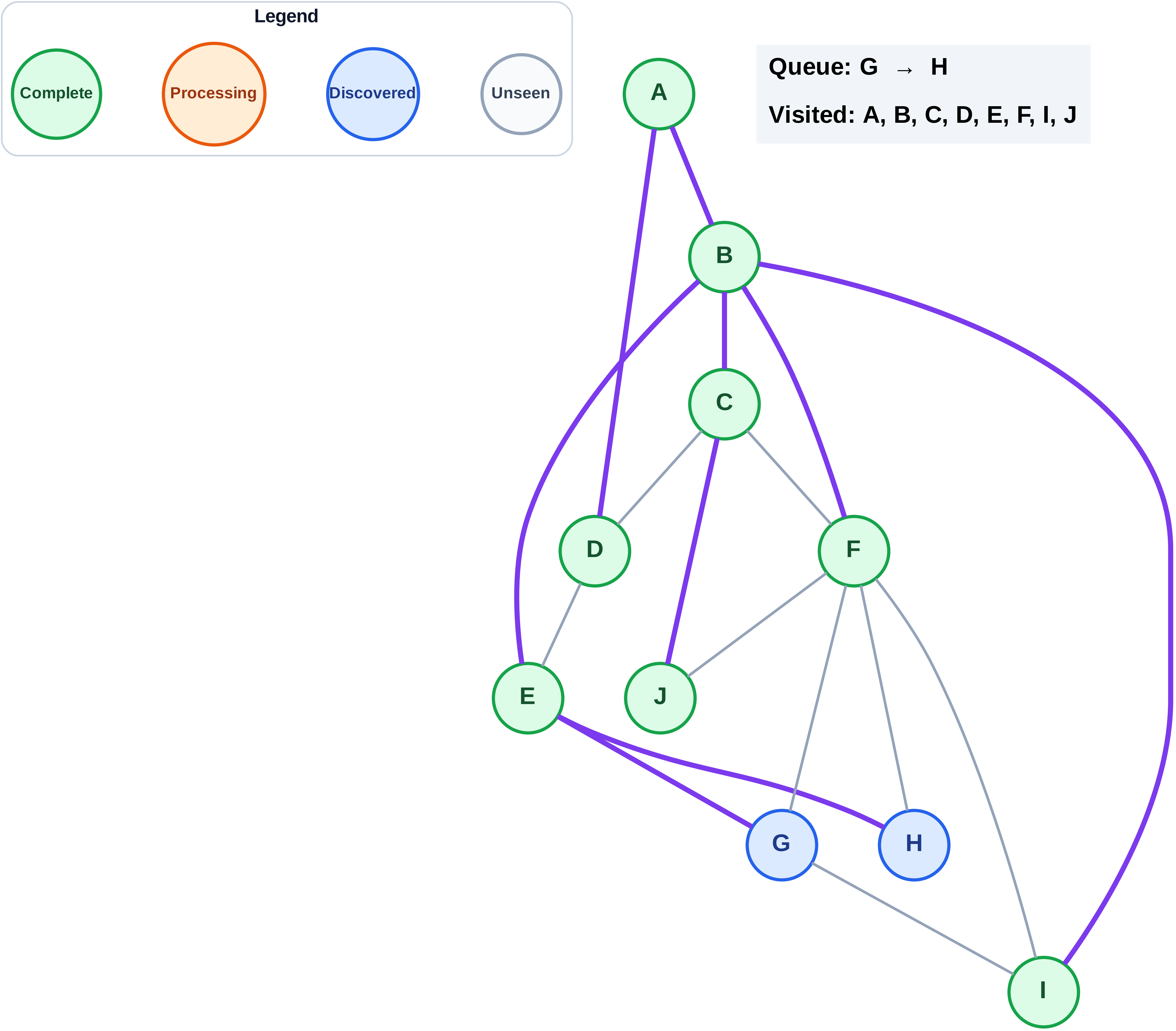
BFS Traversal

Step 16: Process node J



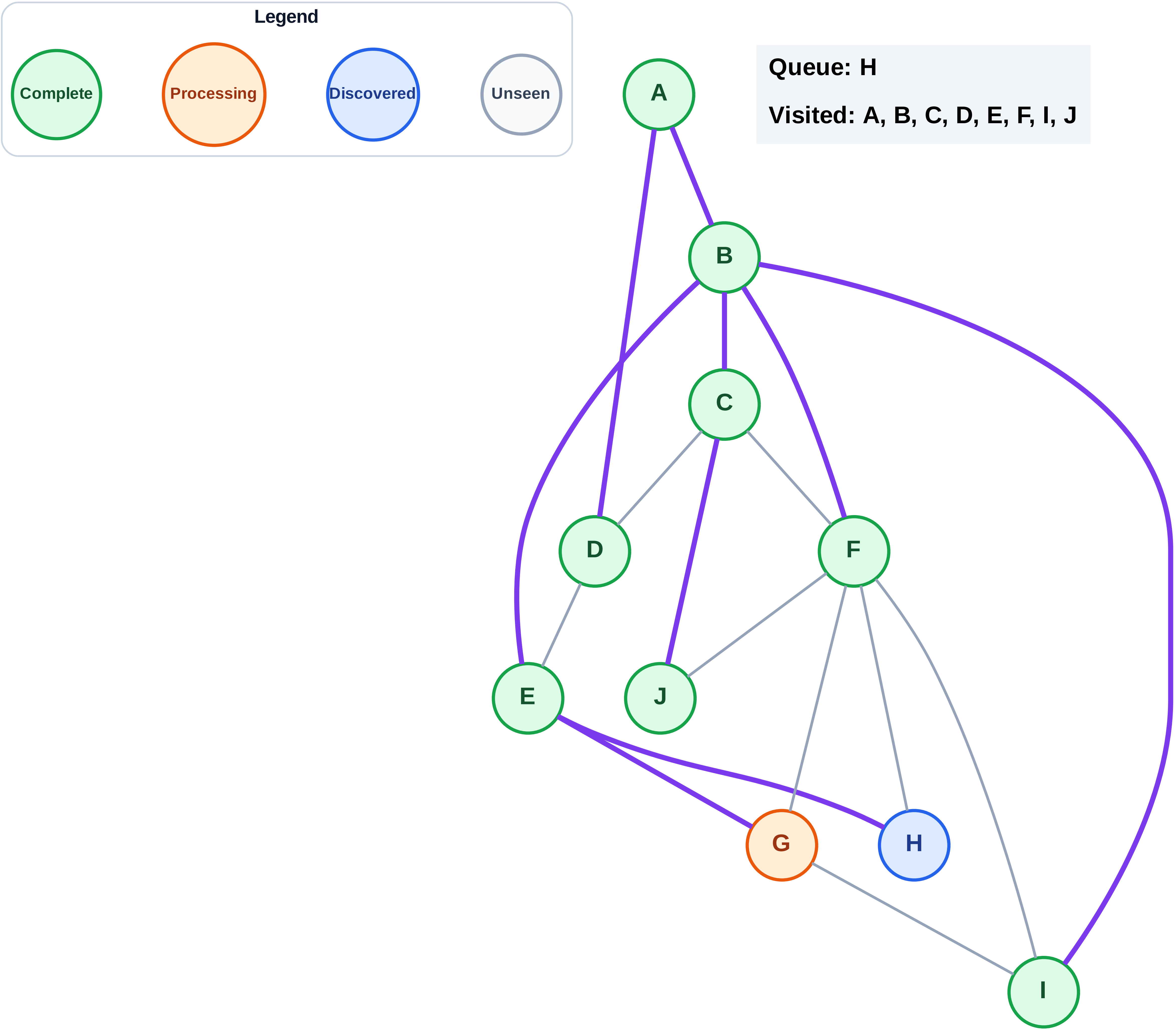
BFS Traversal

Step 17: No new neighbors from J



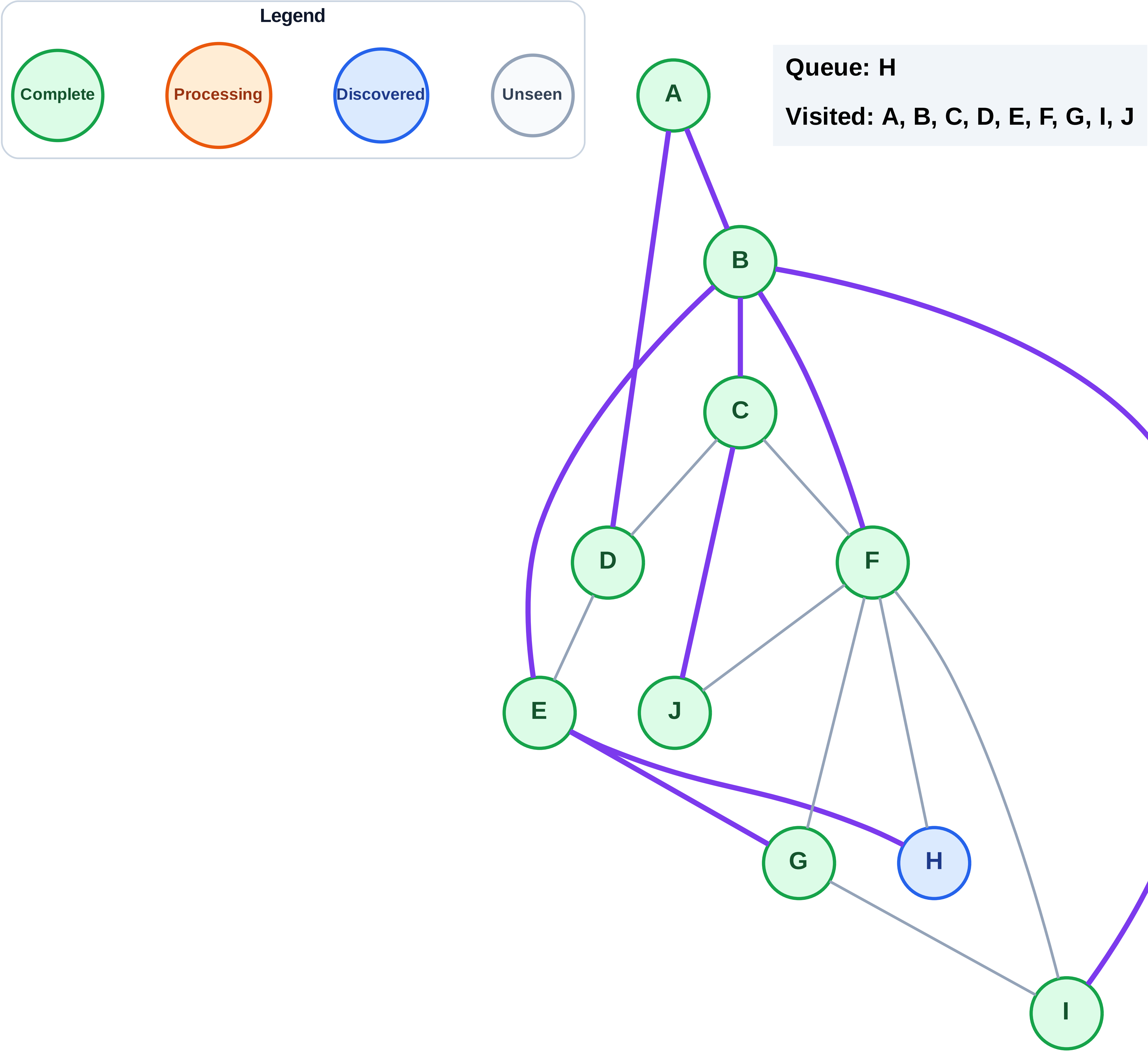
BFS Traversal

Step 18: Process node G



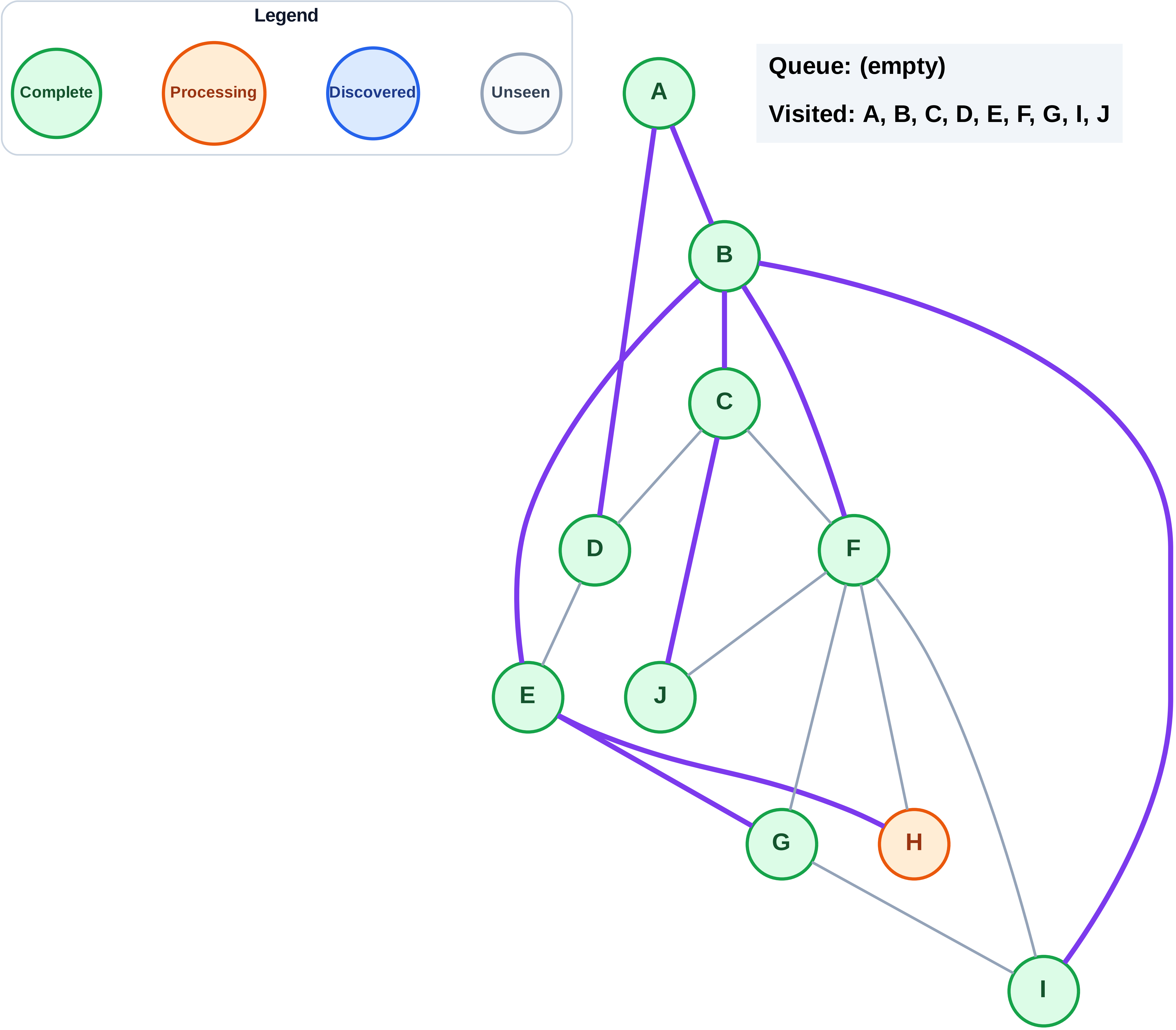
BFS Traversal

Step 19: No new neighbors from G



BFS Traversal

Step 20: Process node H

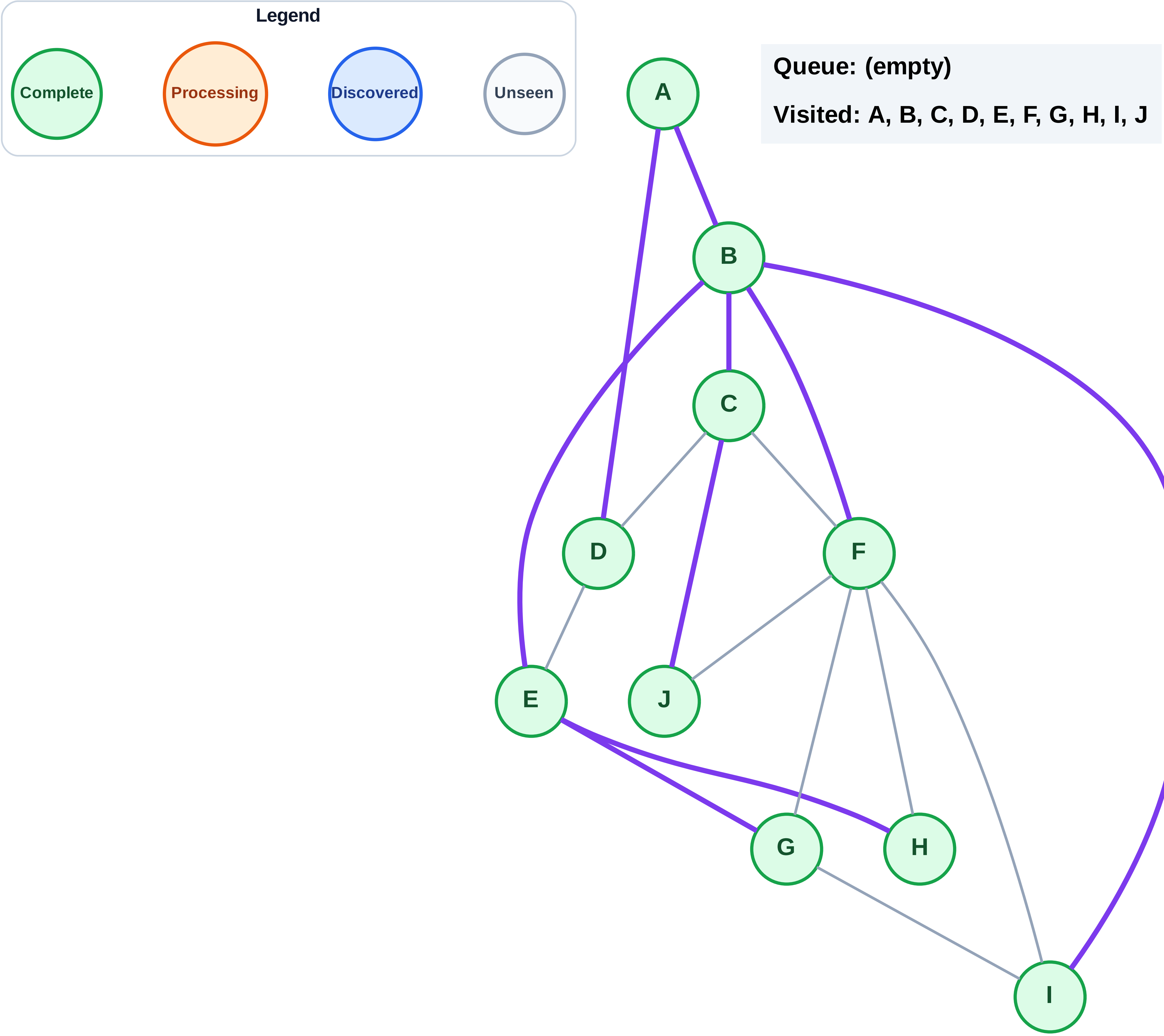


Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J

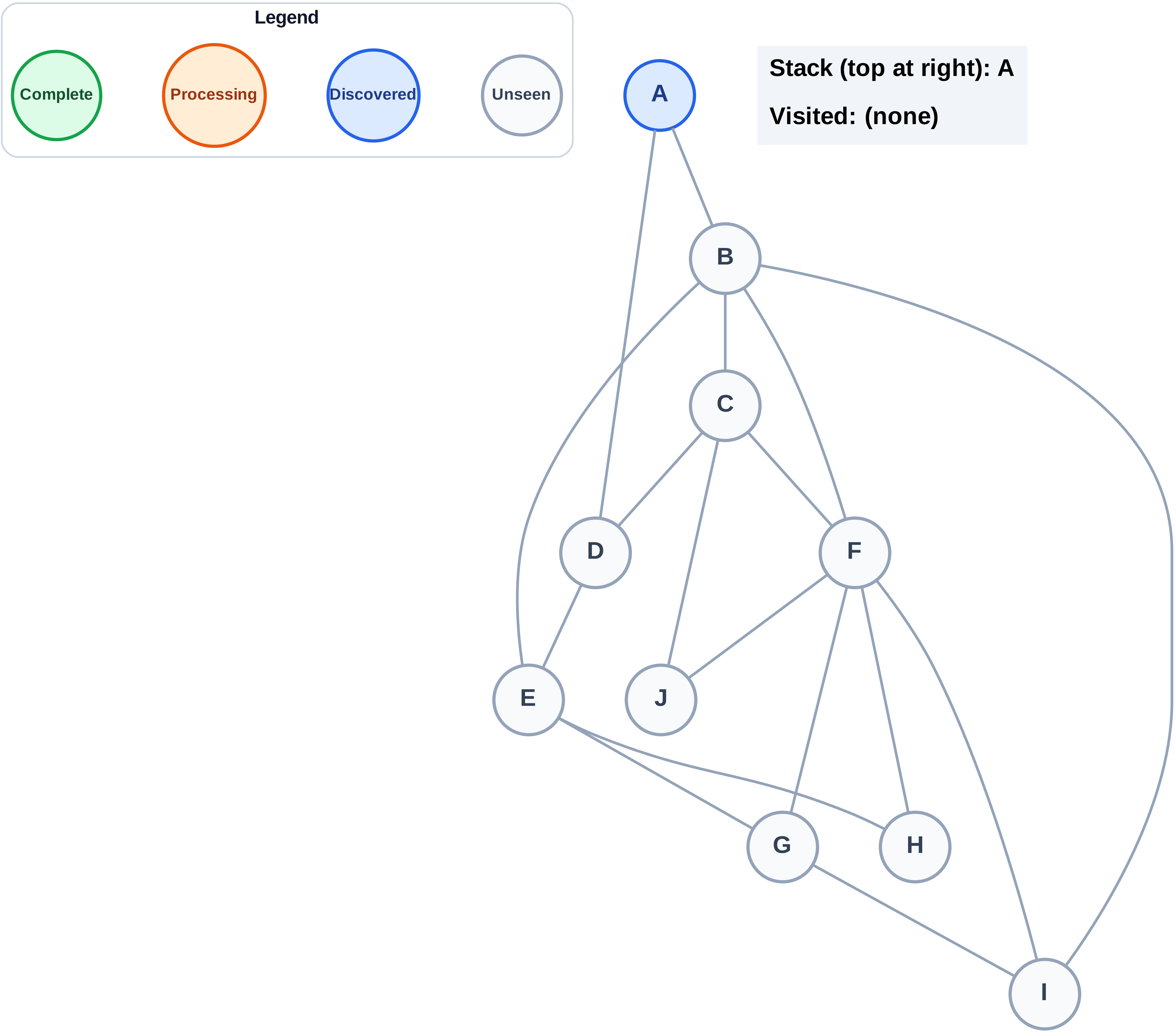
BFS Traversal

Step 21: No new neighbors from H



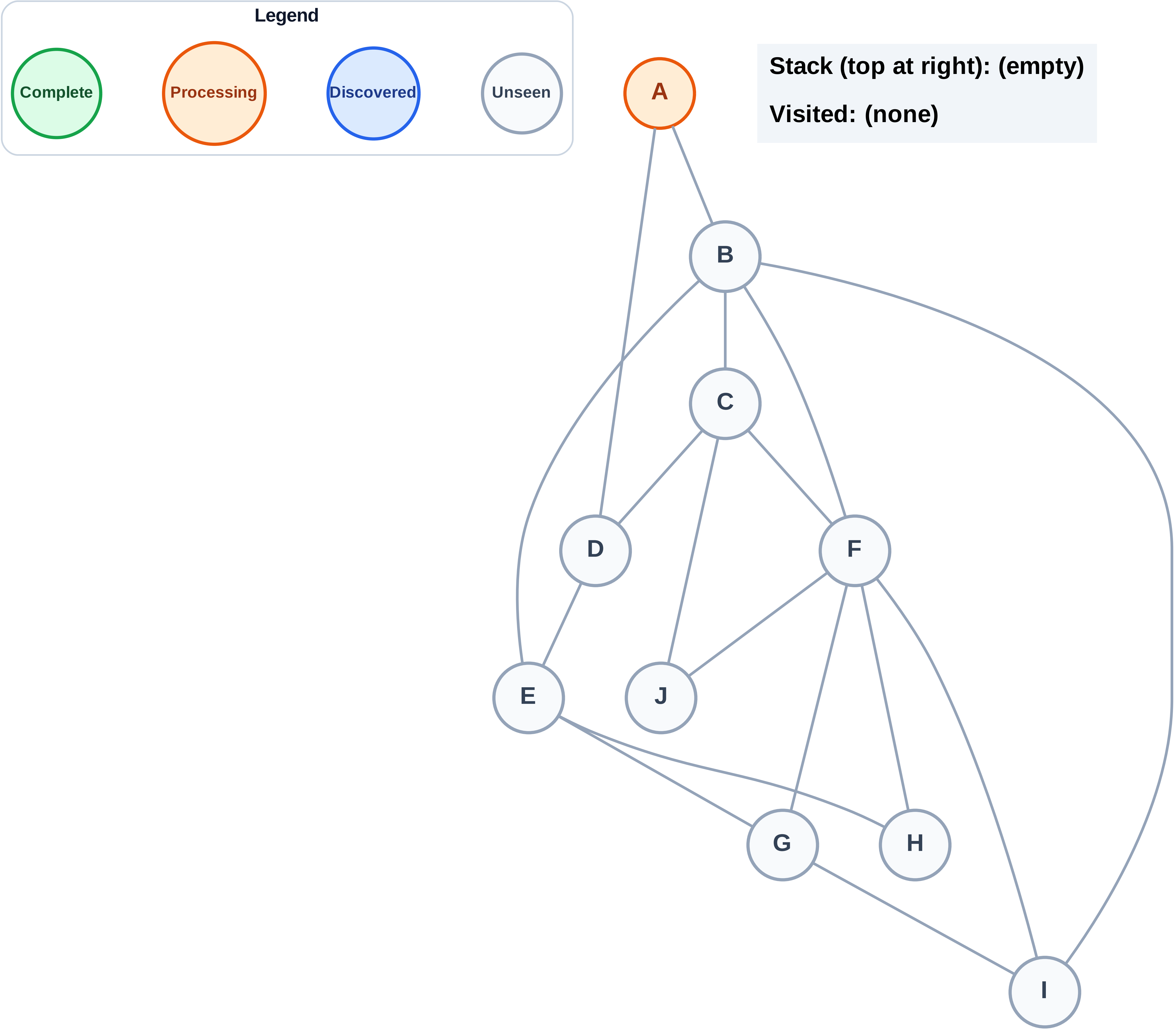
DFS Traversal

Step 1: Start at A



DFS Traversal

Step 2: Process node A

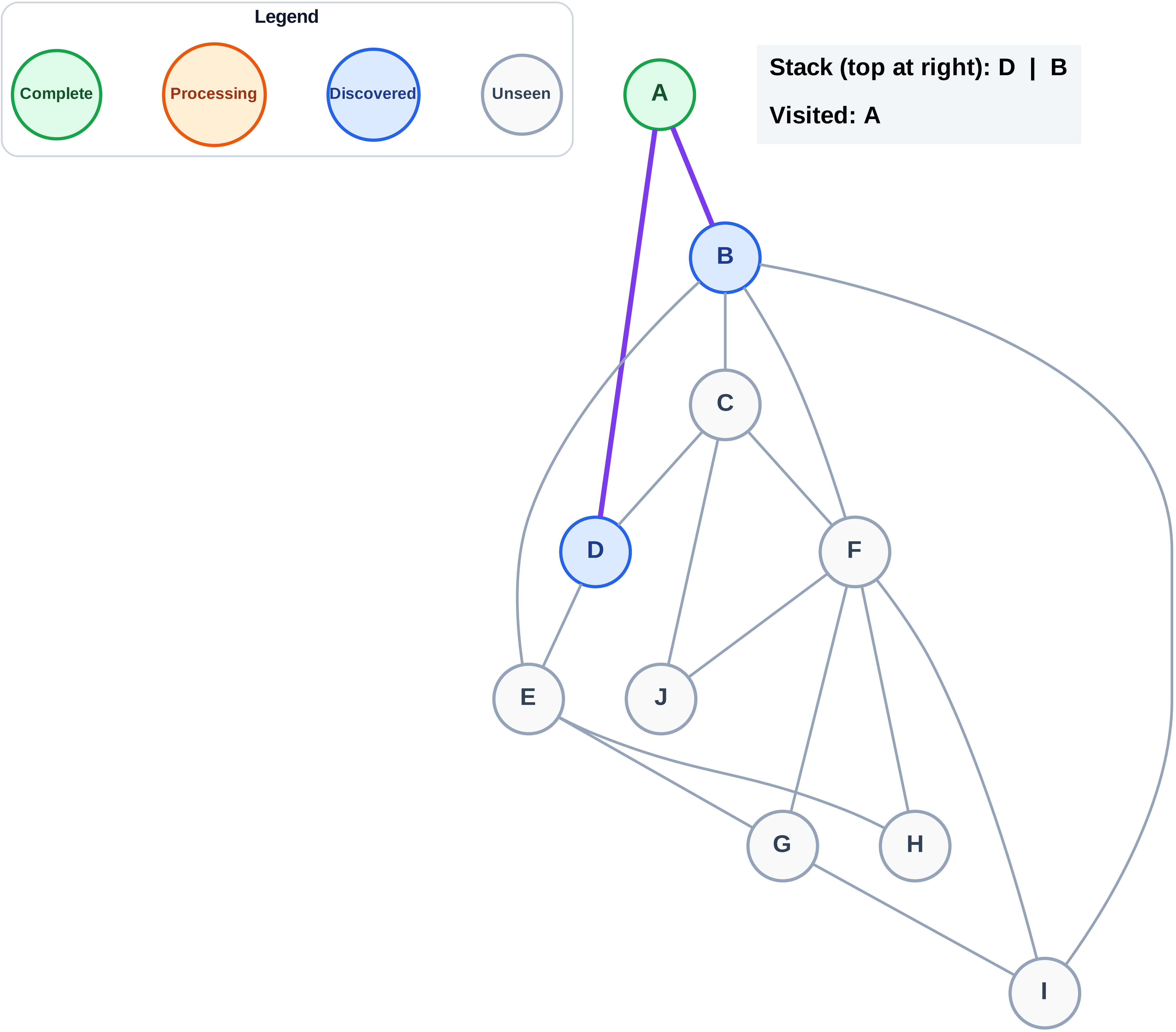


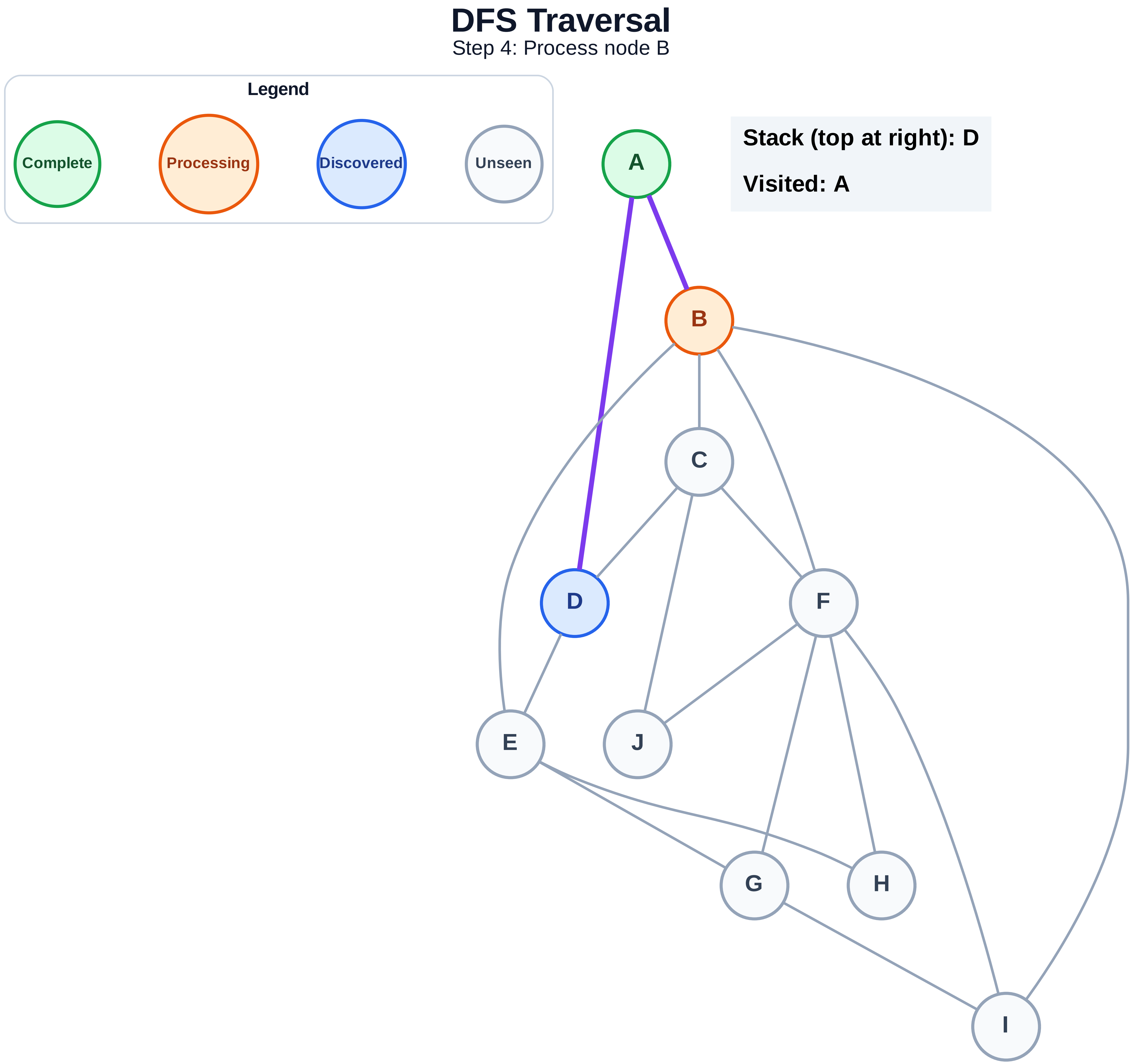
Stack (top at right): (empty)

Visited: (none)

DFS Traversal

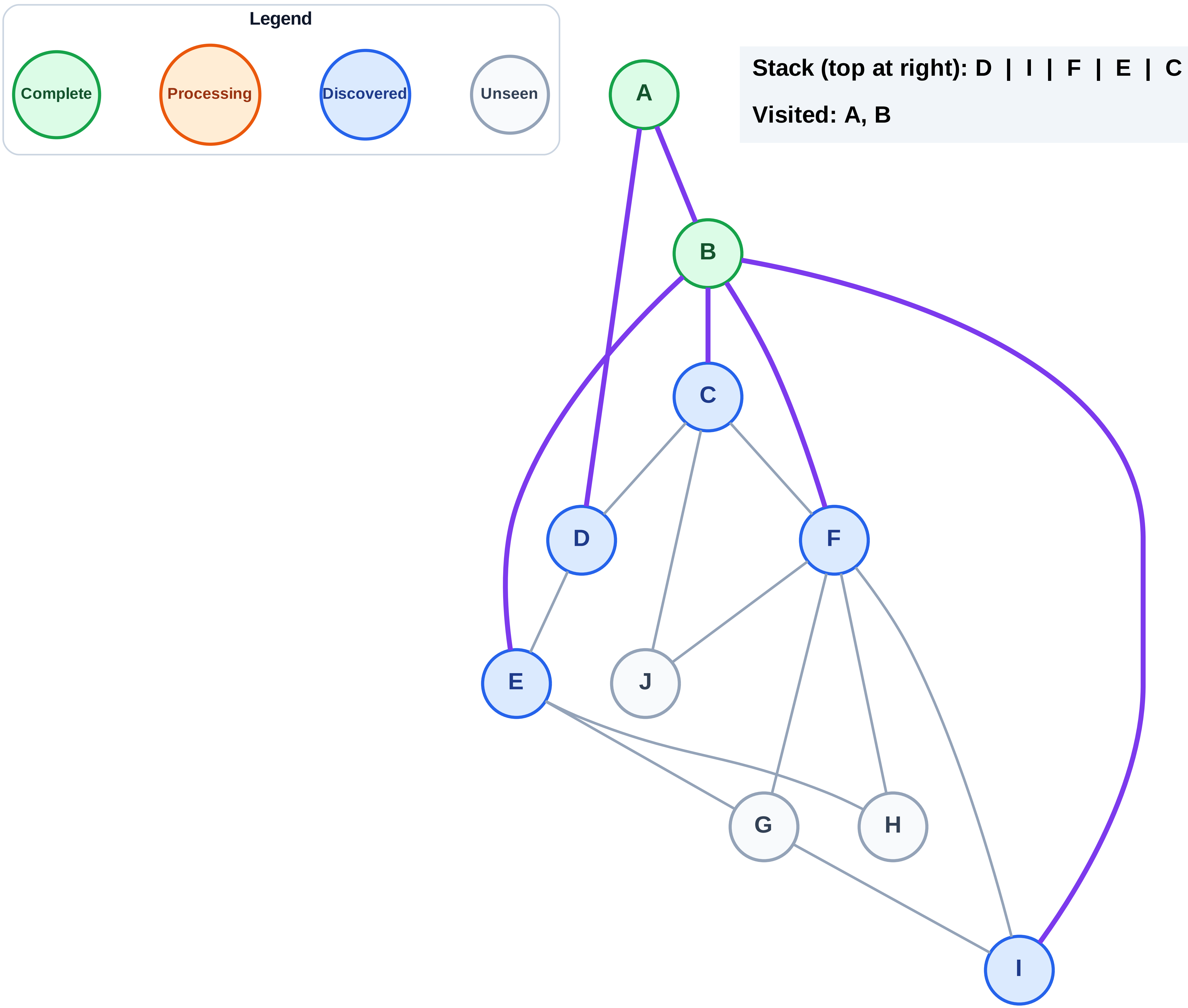
Step 3: Discover D, B from A




DFS Traversal
Step 4: Process node B
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): D
Visited: A
A
B
C
D
F
E
J
G
H
I

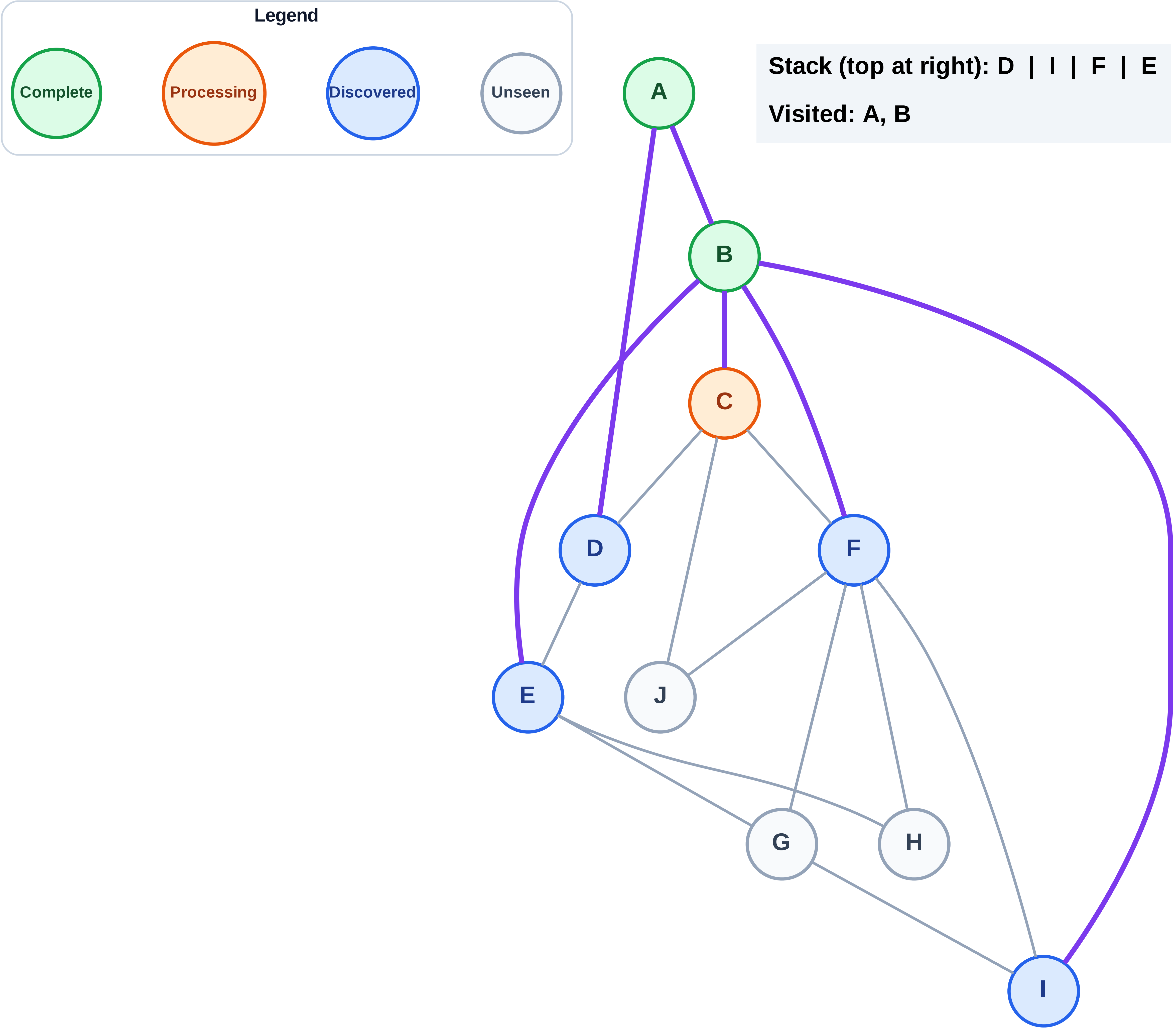
DFS Traversal

Step 5: Discover I, F, E, C from B



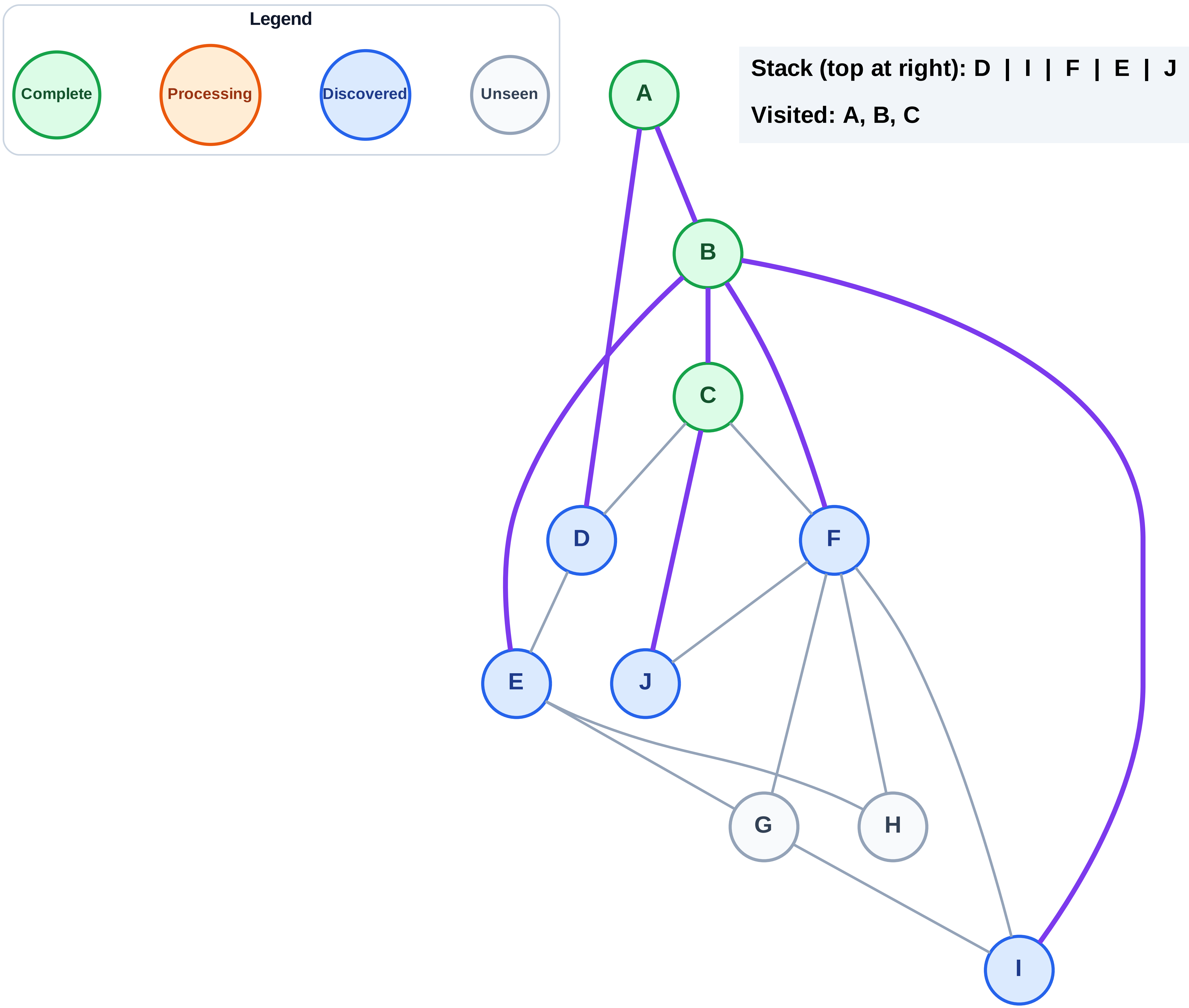
DFS Traversal

Step 6: Process node C



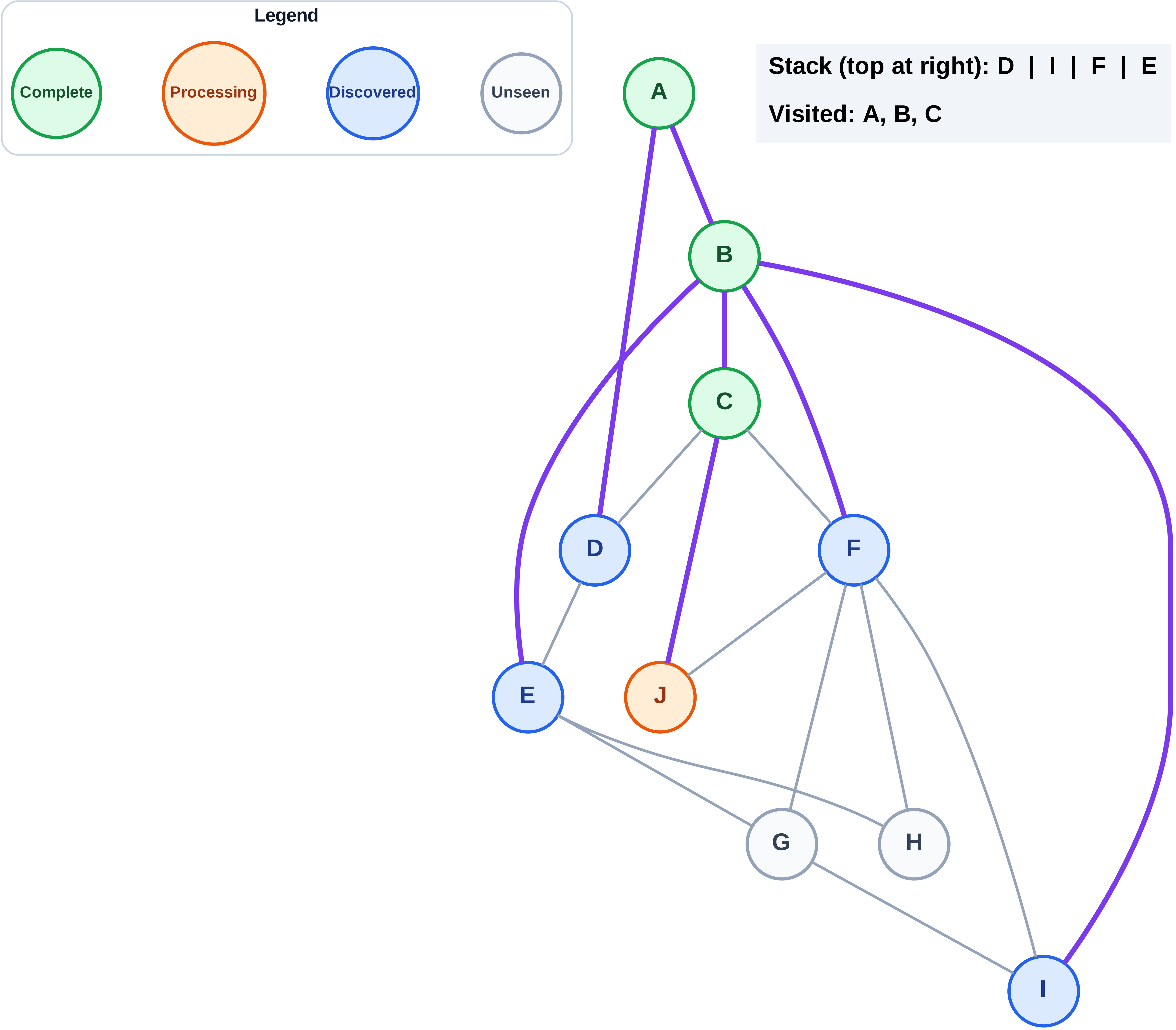
DFS Traversal

Step 7: Discover J from C



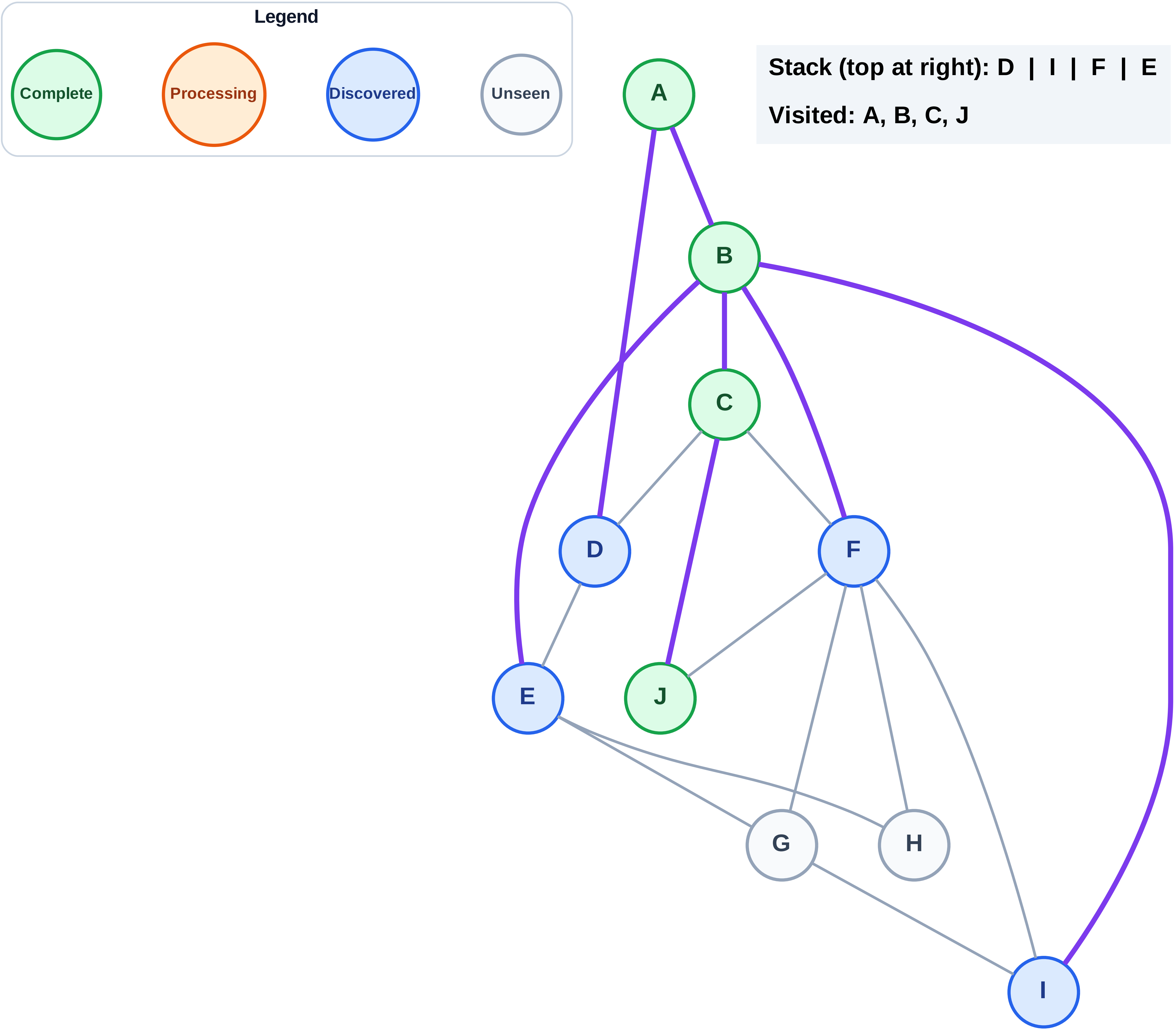
DFS Traversal

Step 8: Process node J



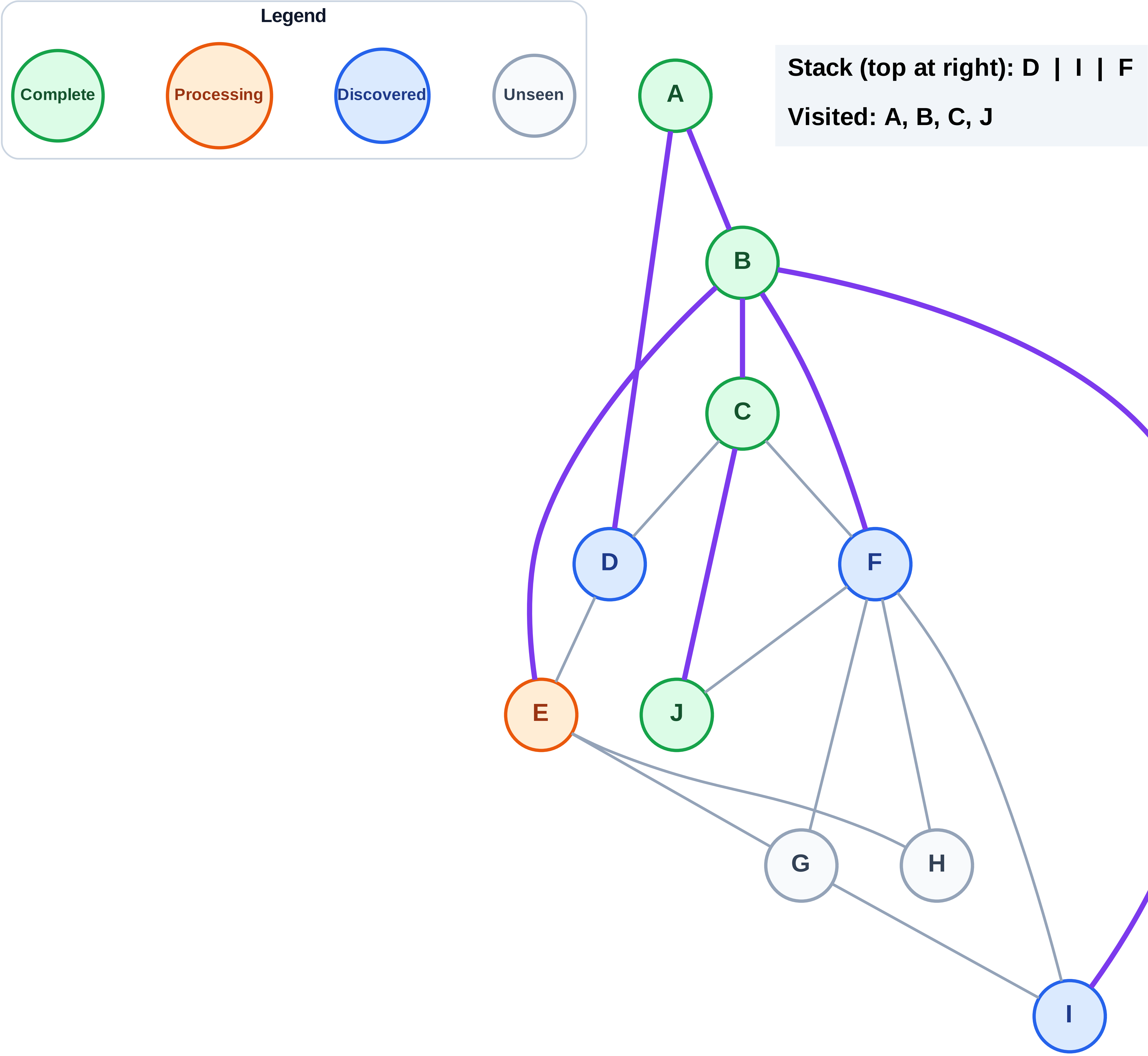
DFS Traversal

Step 9: No new neighbors from J



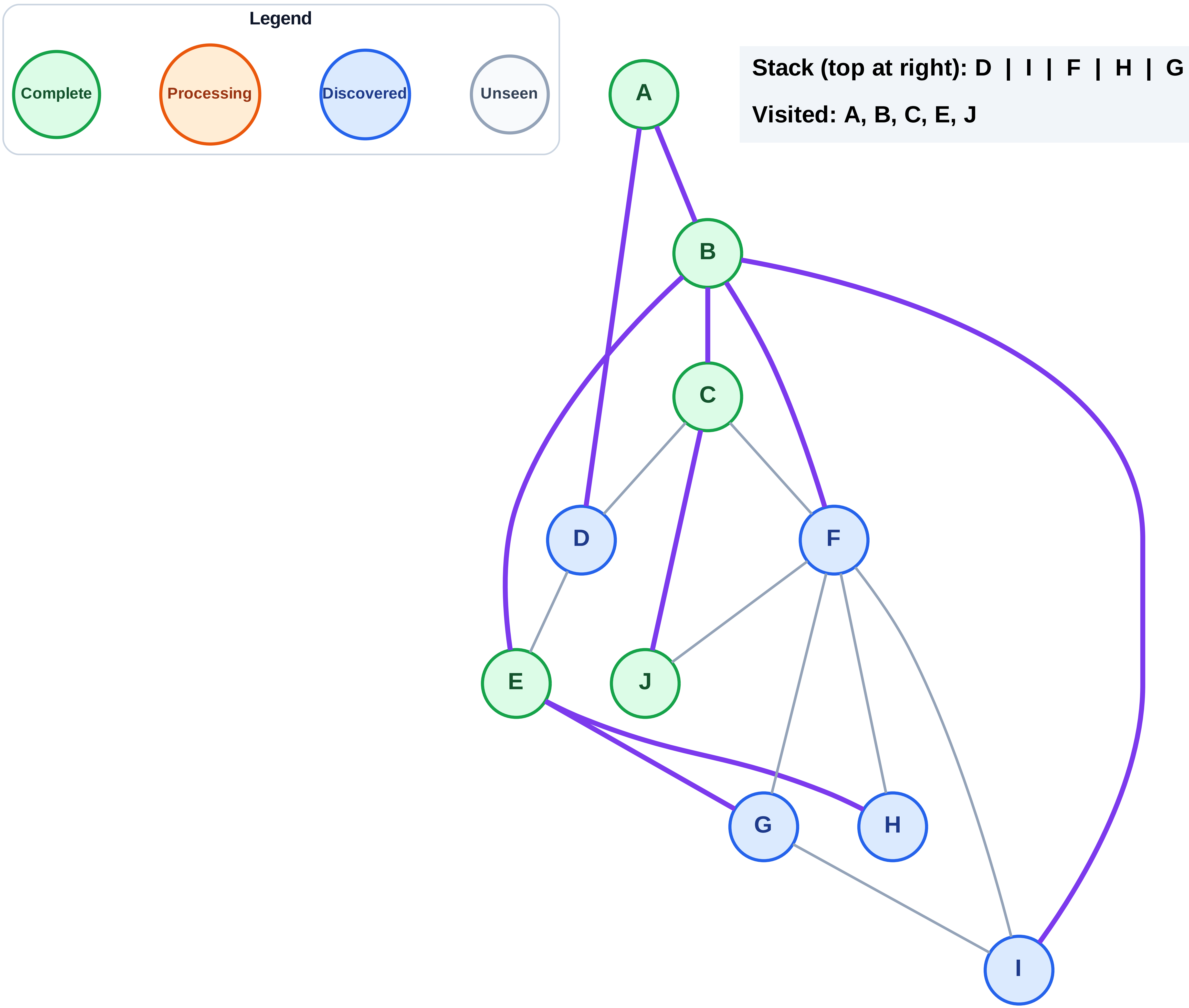
DFS Traversal

Step 10: Process node E



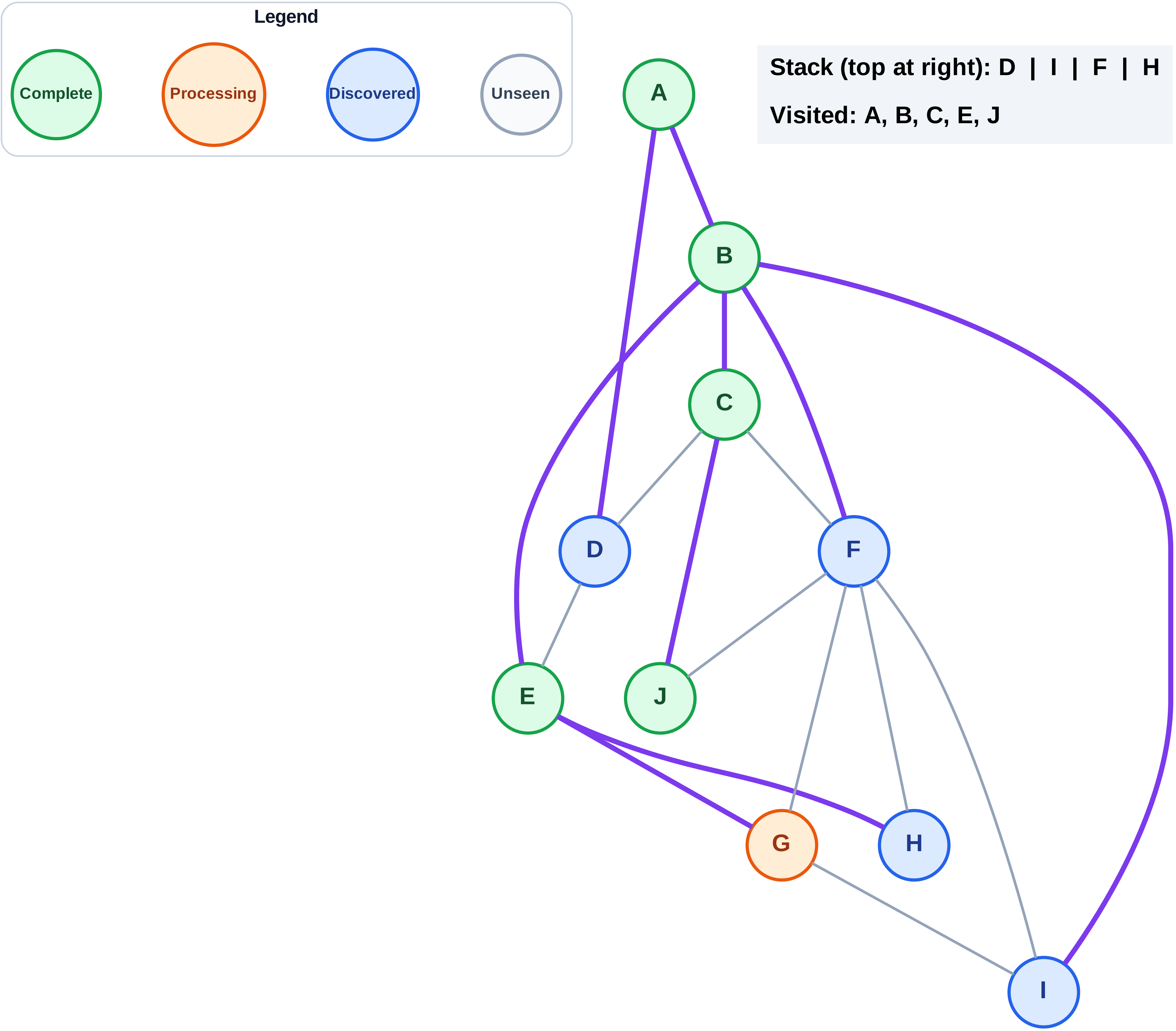
DFS Traversal

Step 11: Discover H, G from E



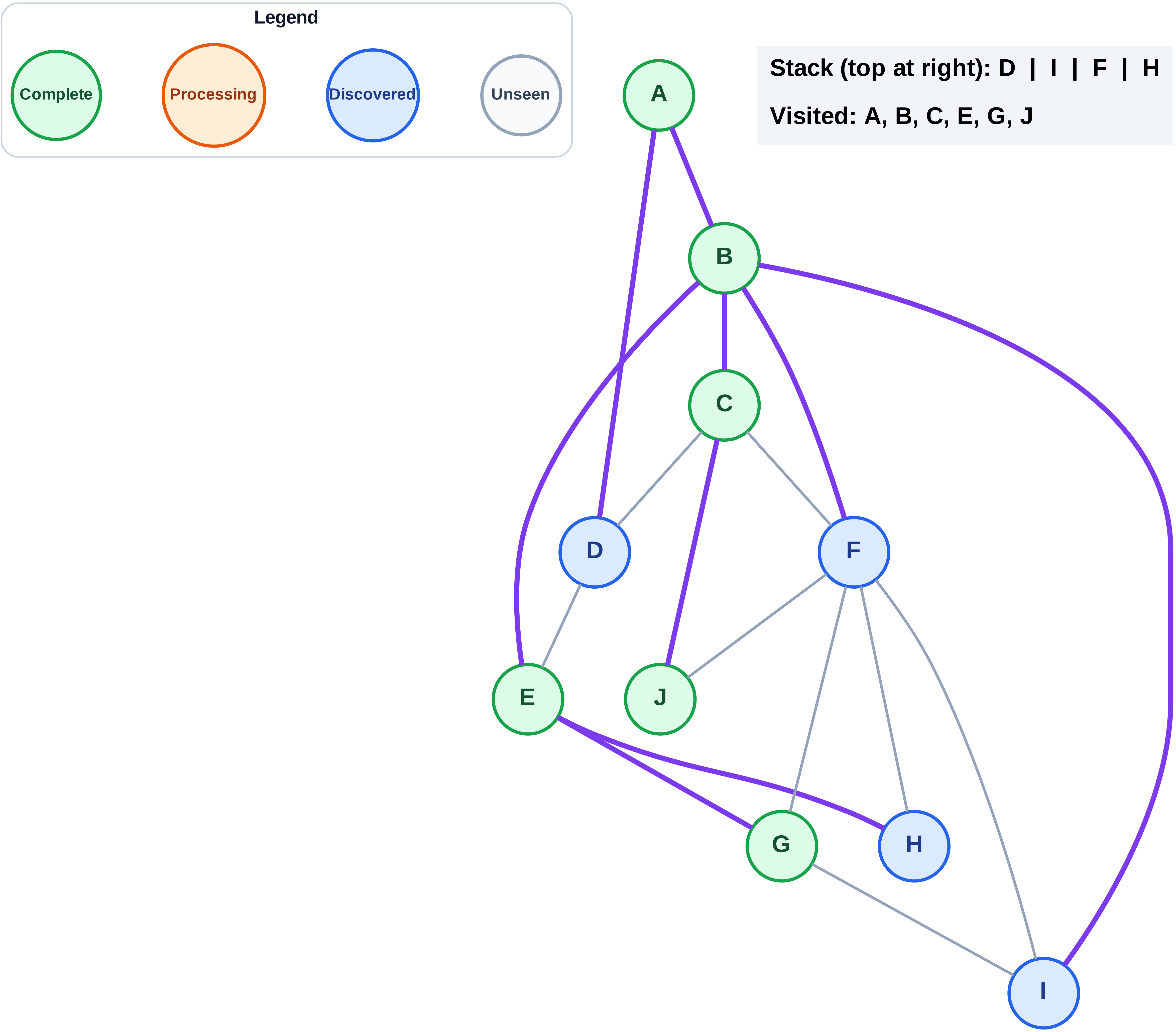
DFS Traversal

Step 12: Process node G



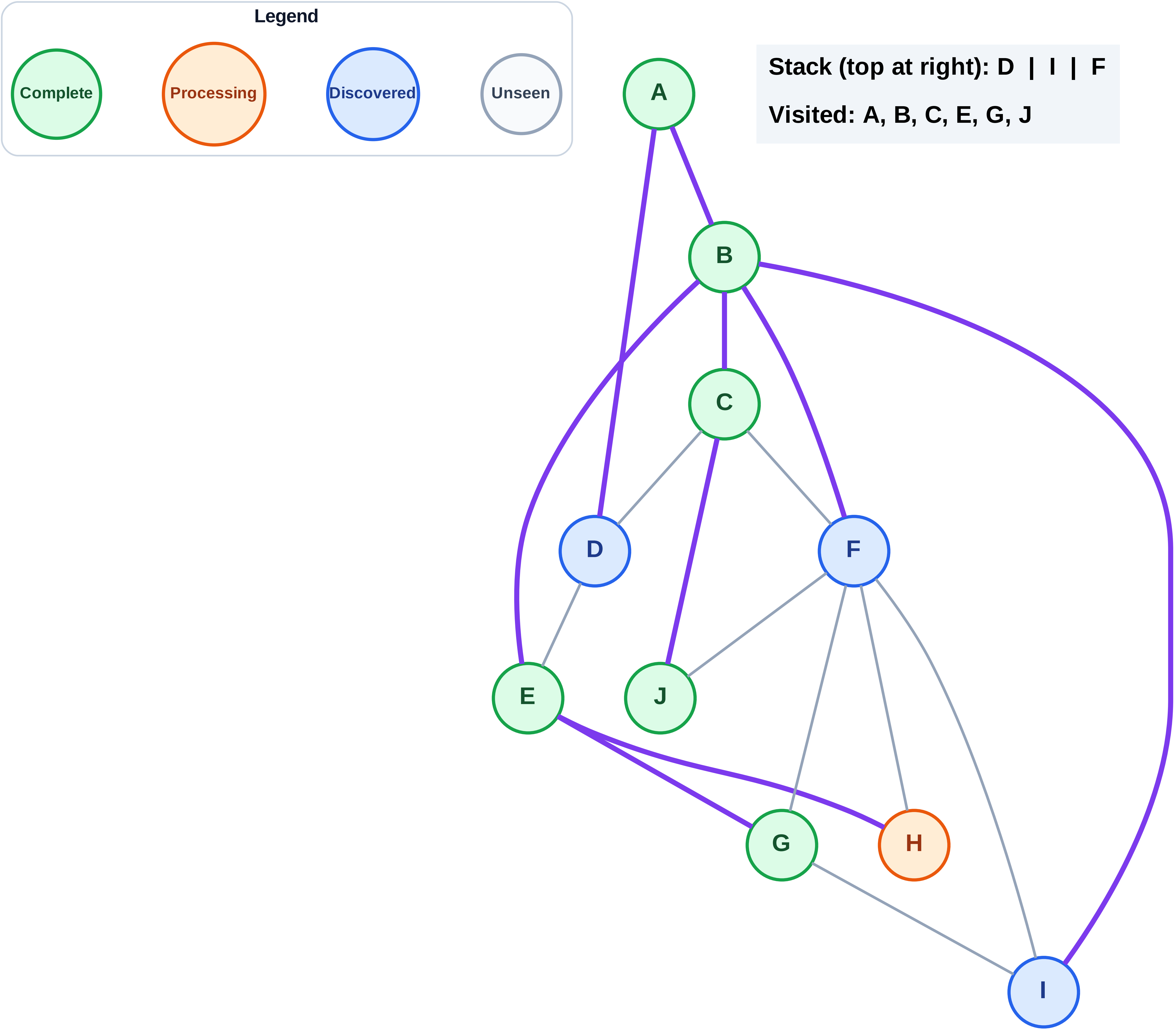
DFS Traversal

Step 13: No new neighbors from G



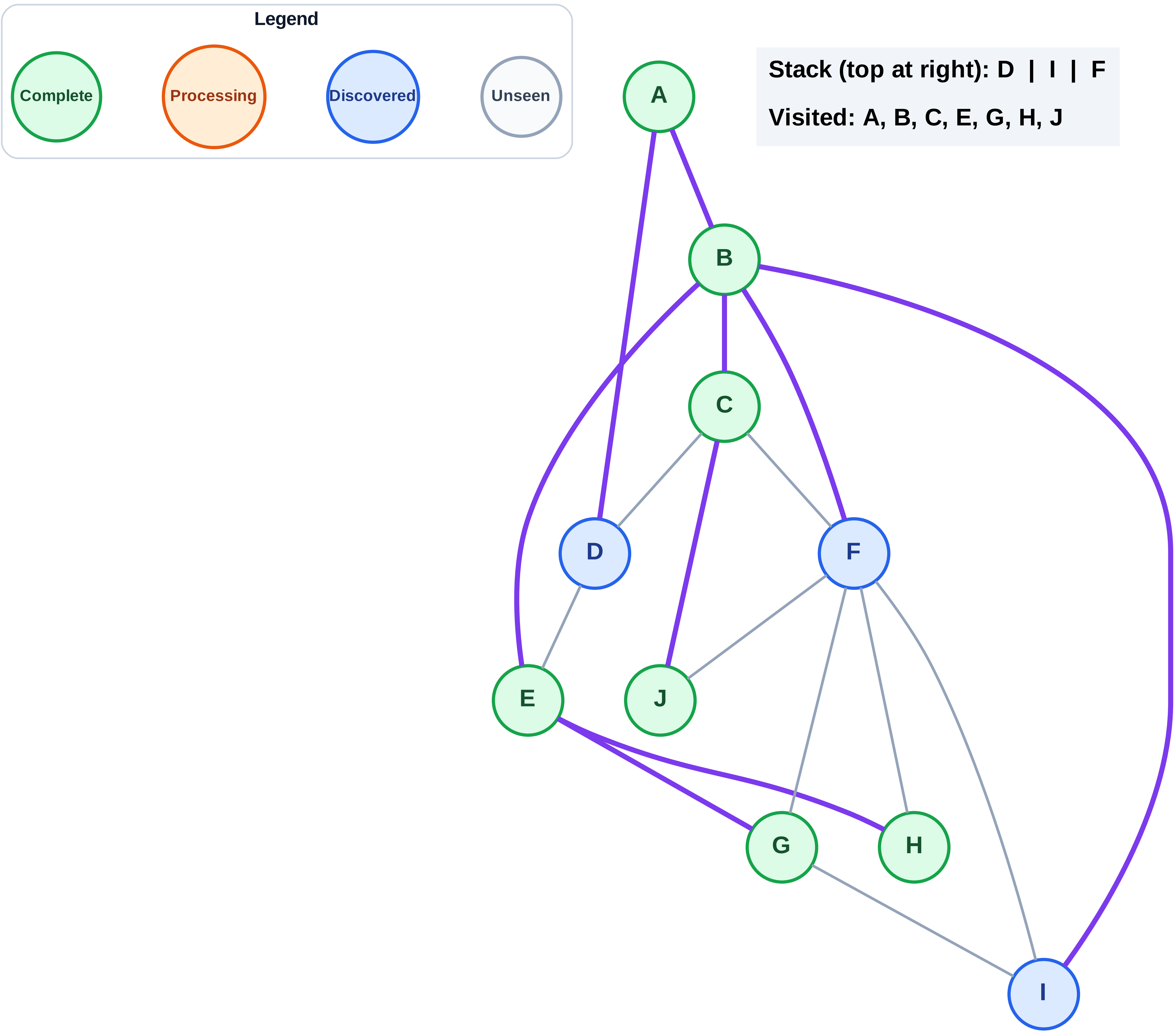
DFS Traversal

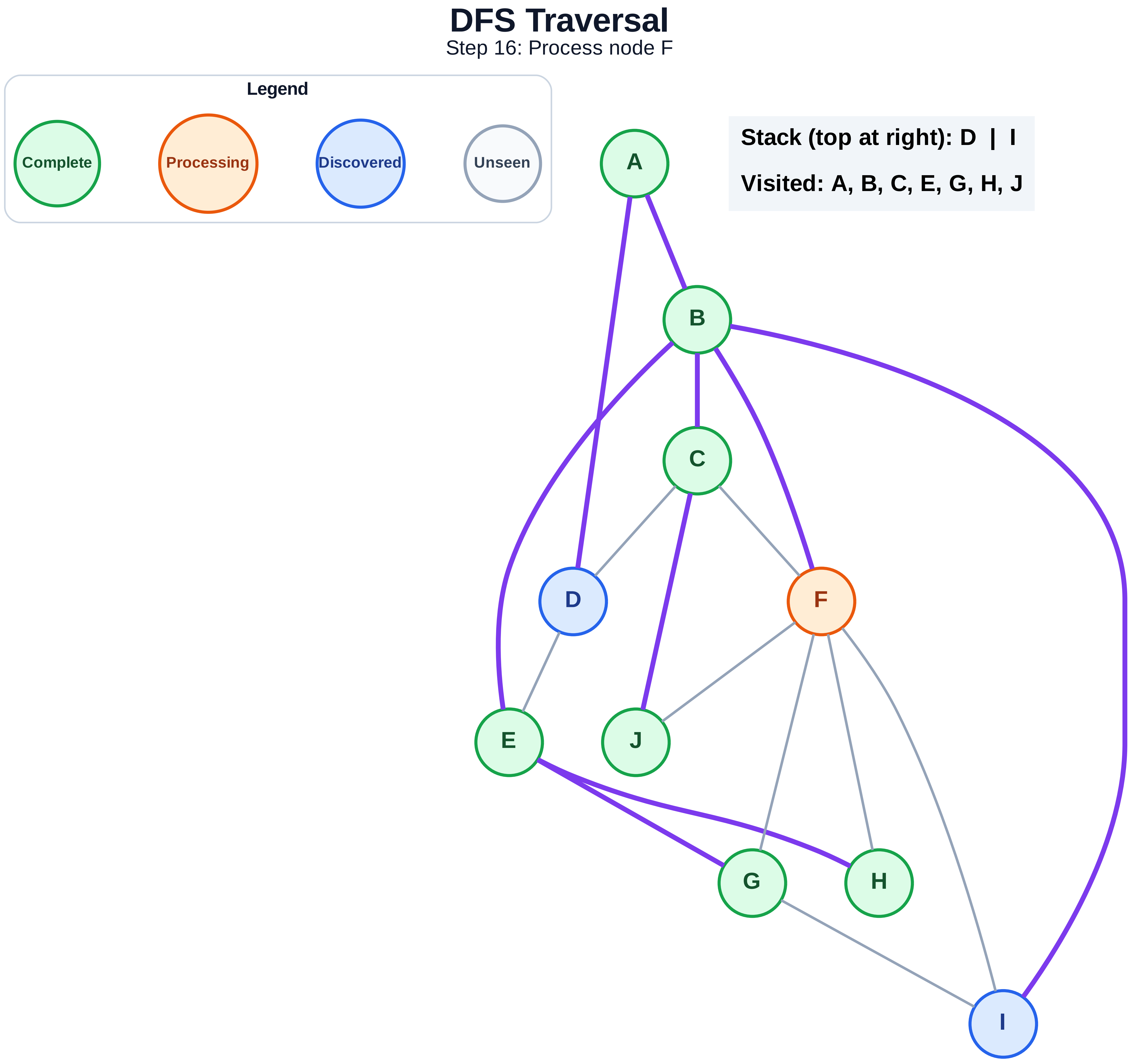
Step 14: Process node H



DFS Traversal

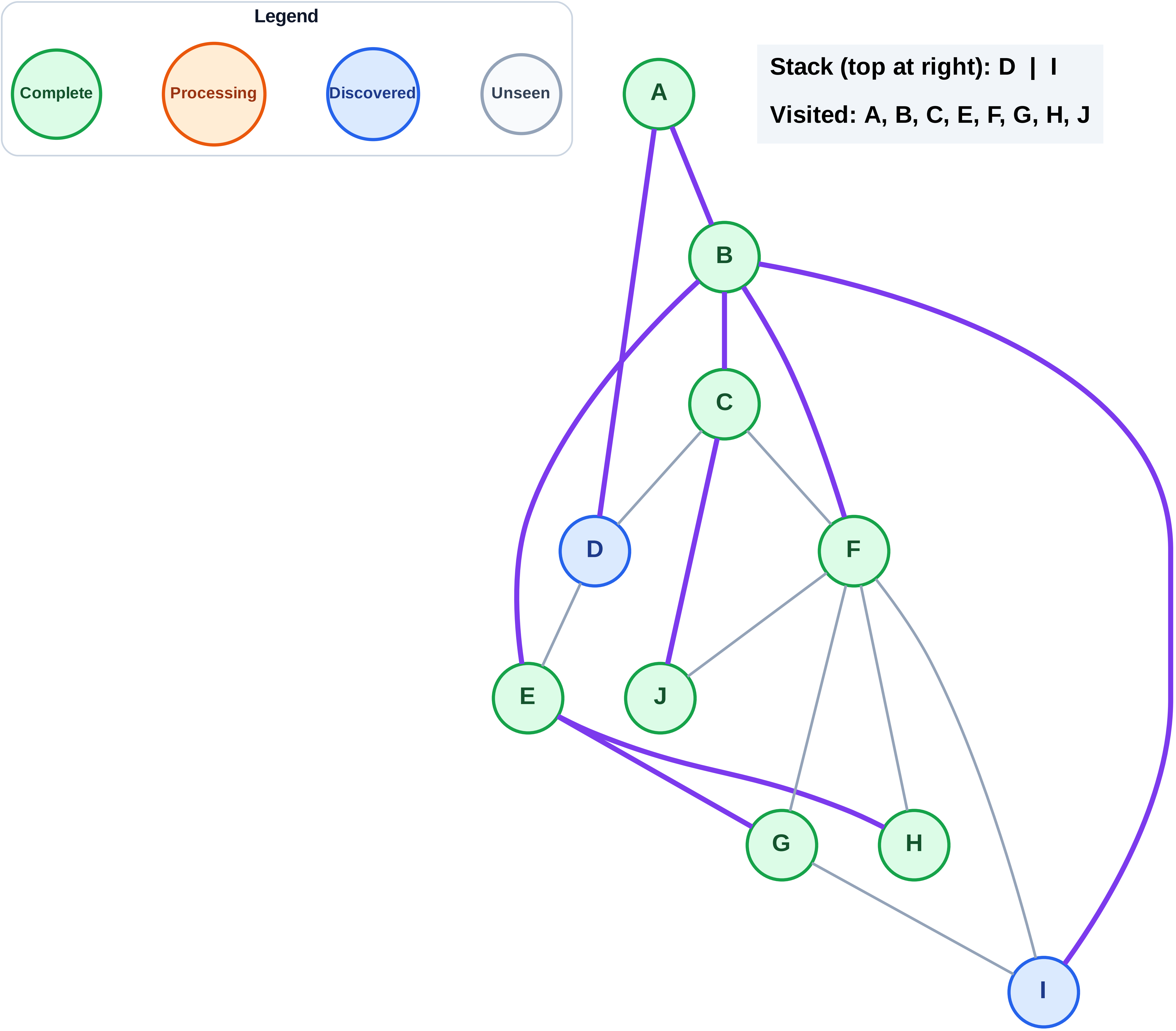
Step 15: No new neighbors from H




DFS Traversal
Step 16: Process node F
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): D | I
Visited: A, B, C, E, G, H, J
A
B
C
D
F
E
J
G
H
I

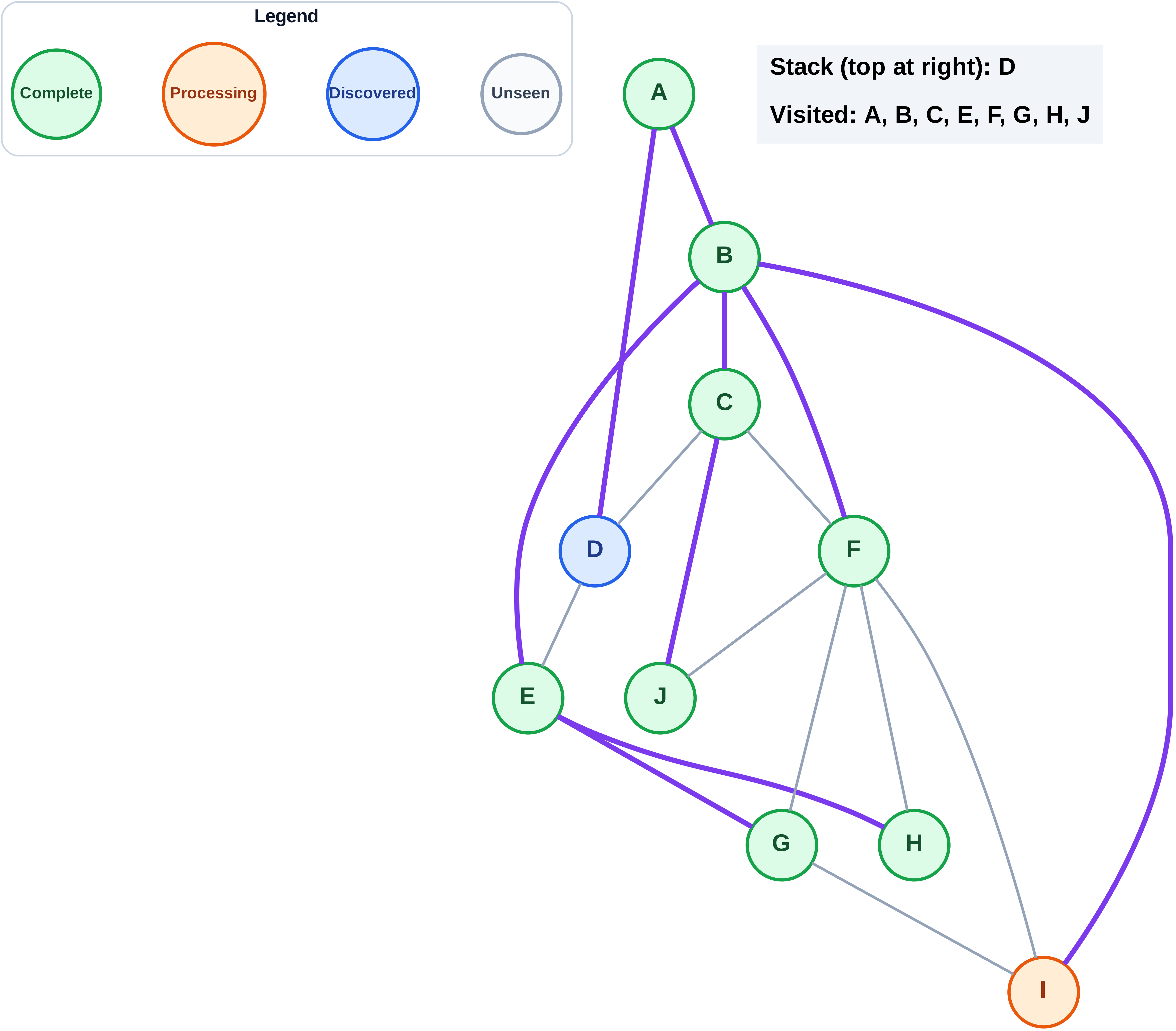
DFS Traversal

Step 17: No new neighbors from F



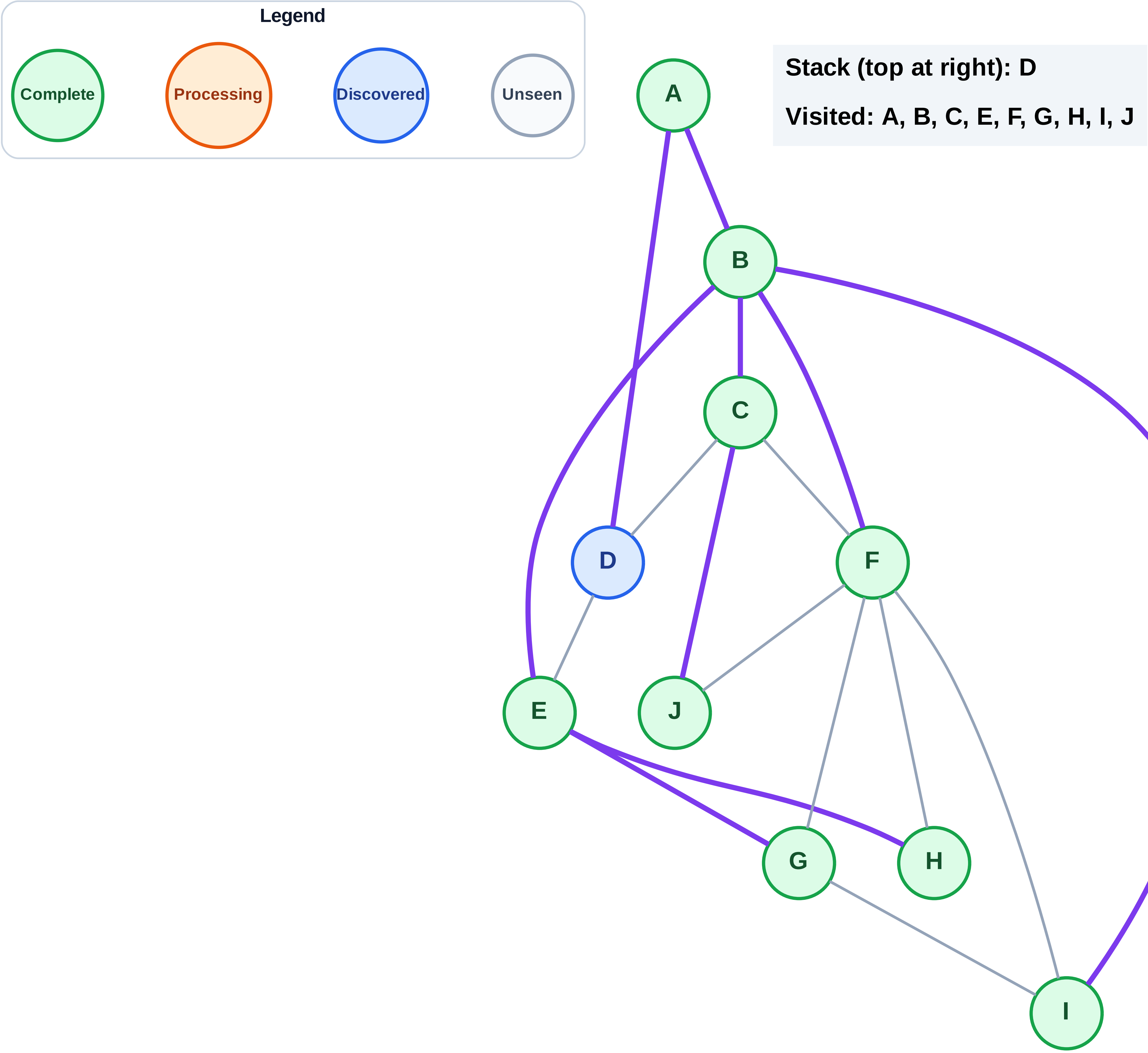
DFS Traversal

Step 18: Process node I



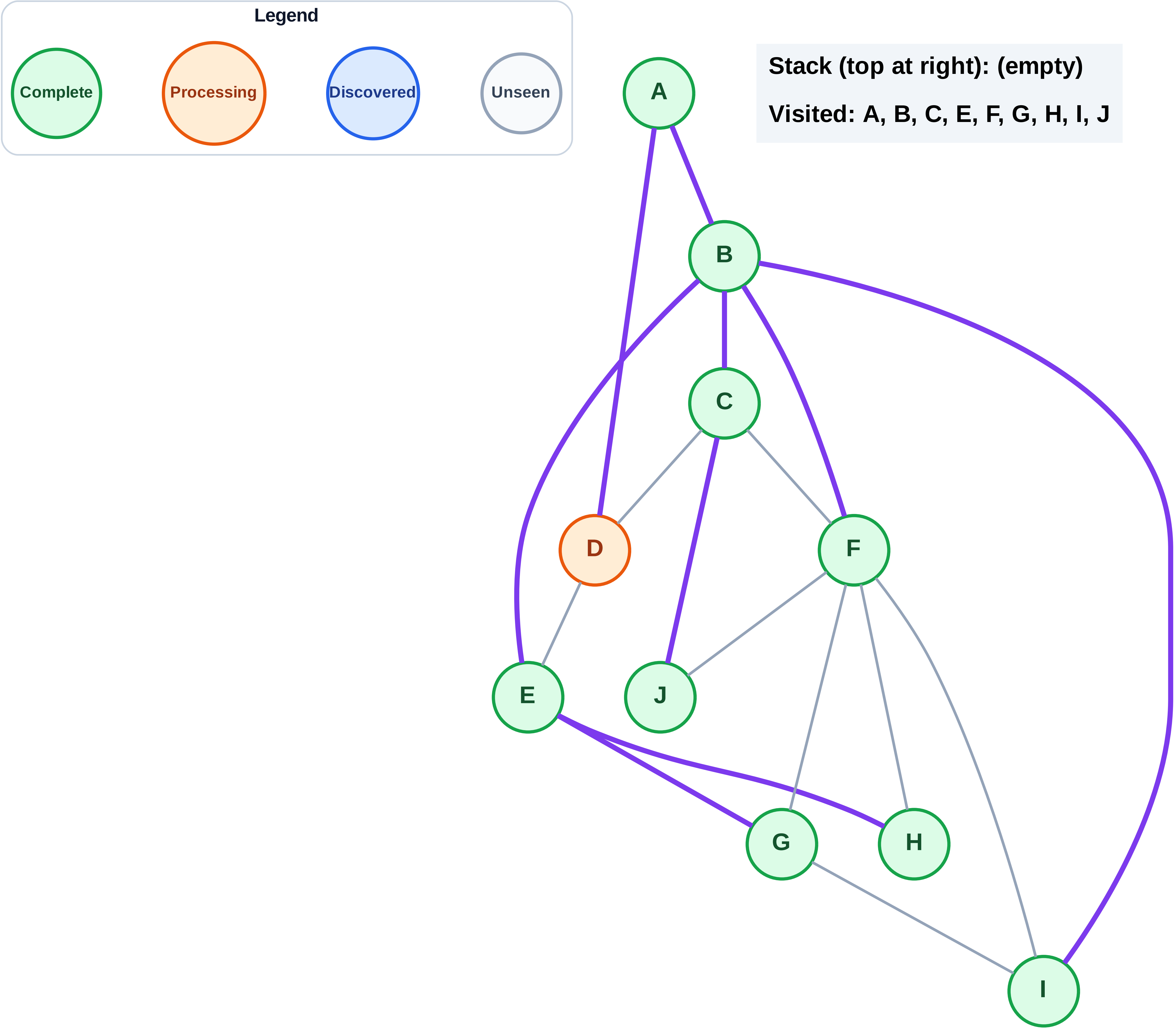
DFS Traversal

Step 19: No new neighbors from I



DFS Traversal

Step 20: Process node D



Stack (top at right): (empty)

Visited: A, B, C, E, F, G, H, I, J

DFS Traversal

Step 21: No new neighbors from D